

Space–Time Algebra

A compact development of paravectors and their physical applications

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Abstract

This document develops a three-generator sigma algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$ and signature $(+, -, -, -)$ are used.

Contents

1	The algebra at a glance	4
2	Hermitian and sigma-conjugate operators	5
3	Paravectors	6
3.1	Core operations	6
3.2	Products and quadratic forms	8
3.3	Quadratic identity and eigenvalues	9
3.4	Real products	9
3.5	Complex products	10
3.5.1	Norms	11
3.5.2	Comparison with the determinant	11
3.6	Projection operators	11
3.6.1	Ladder operators	12
3.7	Functions of paravectors	13
3.7.1	General projector rule	13
3.7.2	Functions of a real unit vector	13
3.7.3	Real paravectors	13
3.7.4	Direct square-root components	14
3.7.5	Complex spectral form	15
3.7.6	Exponential of a general complex paravector	16
3.7.7	Projection derivation	16
3.7.8	Direct series derivation	17

3.7.9	Component form and real/imaginary limits	17
3.7.10	Kinematic form	18
3.7.11	Cayley transform	18
3.7.12	Aligned boost and rotation	19
3.7.13	Logarithm of a general complex paravector	20
3.7.14	Special cases	20
4	Transformations and invariants	21
4.1	Similarity invariants	21
4.1.1	Trace invariance	21
4.1.2	Determinant invariance	22
4.1.3	Norm invariance	22
5	Differentiated fields	23
5.1	Paravector differential operators	23
5.2	Differentiated paravector fields	23
6	Space–time kinematics	25
6.1	Intervals, velocity, and momentum	25
6.1.1	Proper acceleration	27
6.2	Varying transformations and second differentials	29
6.3	Free-particle action	30
7	Electromagnetism	31
7.1	Field equation and potentials	31
7.2	Field invariants	33
7.3	Lorentz force from the field and action	33
7.4	Proper acceleration from the Lorentz force	35
7.5	Energy balance	36
8	Relativistic wave equations	37
8.1	Paravector inner products and expectations	38
8.2	Physical spinors and minimal ideals	39
8.3	Spin carried by paravector components	41
8.4	Energy and momentum as differential operators	42
8.5	Klein–Gordon equation	42
8.6	Spin-zero low-energy limit	43
8.7	Dirac equation in block form	44

8.8	Schrödinger–Pauli limit	46
8.9	Interacting operators and factorization	47
8.10	Spin, energy sign, and probabilities	48
9	Rotations and Lorentz boosts	51
9.1	Rotors	51
9.2	Boosts	52
10	Flat spacetime in generalized coordinates	53
A	Pauli-matrix representation	54
A.1	Faithful representation	55
A.2	Matrix operations and induced algebra operations	55
A.2.1	Trace, determinant, and adjugate	57
A.3	Algebra–matrix crosswalk	58
A.4	Matrix amplitudes and column spinors	58
A.5	Quaternion subalgebra	61
B	Reference summary	63
C	Conventions and cautions	68

1 The algebra at a glance

The generators σ_n are algebraic objects with ordered multiplication rules. They are not ordinary vector components: their products generally do not commute, so the order of the factors is part of every expression. The following panel previews the generators, their product, and the general complex element developed below.

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations use

$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3, \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \quad (1.2)$$

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (1.3)$$

Thus the square of a vector always denotes its self dot product.

The set $\{1, \boldsymbol{\sigma}, \mathbf{i}\boldsymbol{\sigma}, \mathbf{i}\}$ is a complete basis, displayed by type as

$$\underbrace{1}_{\text{scalar}}, \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}}, \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}}, \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

Anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. The pseudoscalar is central: moving σ_n through the other two generators produces two sign changes, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}. \quad (1.4)$$

Thus its order of multiplication does not matter.

The commutator is

$$[X, Y] := XY - YX \quad (1.5)$$

$$[Y, X] = -[X, Y]. \quad (1.6)$$

Thus $[X, Y] = 0$ exactly when X and Y commute. For $n, m, k \in \{1, 2, 3\}$, define the Kronecker delta by

$$\delta_{nm} := \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \quad (1.7)$$

and the Levi-Civita symbol by

$$\epsilon_{nmk} := \begin{cases} +1, & (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two or more indices are equal.} \end{cases} \quad (1.8)$$

For $n, m \in \{1, 2, 3\}$, the ordered sigma multiplication rule is

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9)$$

$$[\sigma_n, \sigma_m] = 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k. \quad (1.10)$$

2 Hermitian and sigma-conjugate operators

Define the two operations by

$$(\cdot)^{\mathbf{H}} := \text{reverse the order of every product of sigma generators}, \quad (2.1)$$

$$(\cdot)^* := \text{replace every } \sigma_n \text{ by } -\sigma_n \text{ without changing product order}. \quad (2.2)$$

Thus, for two generators,

$$(\sigma_m \sigma_n)^{\mathbf{H}} = \sigma_n \sigma_m, \quad (2.3)$$

$$(\sigma_m \sigma_n)^* = (-\sigma_m)(-\sigma_n). \quad (2.4)$$

Both operations act term by term on sums. Their actions on the four basis types are

type representative	scalar 1	vector σ_n	bivector $\mathbf{i} \sigma_n$	pseudoscalar $\mathbf{i}$
$(\cdot)^*$	+	−	+	−
$(\cdot)^{\mathbf{H}}$	+	+	−	−
$(\cdot)^{* \mathbf{H}}$	+	−	−	+

In particular,

$$\sigma_n^* = -\sigma_n, \quad (2.5)$$

$$\sigma_n^{\mathbf{H}} = \sigma_n, \quad (2.6)$$

$$\mathbf{i}^* = -\mathbf{i}, \quad (2.7)$$

$$\mathbf{i}^{\mathbf{H}} = -\mathbf{i}. \quad (2.8)$$

For the bivectors, these actions combine to give

$$(\mathbf{i} \sigma_n)^* = \mathbf{i}^* \sigma_n^* = \mathbf{i} \sigma_n \quad (2.9)$$

$$(\mathbf{i} \sigma_n)^{\mathbf{H}} = \sigma_n^{\mathbf{H}} \mathbf{i}^{\mathbf{H}} = -\mathbf{i} \sigma_n. \quad (2.10)$$

Thus sigma conjugation preserves the original order of the factors, whereas the Hermitian operator changes their order. For arbitrary algebra elements Z, Z' , these definitions give

$$(ZZ')^{\mathbf{H}} = Z'^{\mathbf{H}} Z^{\mathbf{H}}, \quad (ZZ')^* = Z^* Z'^*, \quad Z^{* \mathbf{H}} = Z^{\mathbf{H} *}.$$

Appendix A reproduces this as Hermitian conjugation in the Pauli-matrix representation.

3 Paravectors

Throughout this section, a general complex paravector is $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}$ unless a subsection states a real specialization.

3.1 Core operations

Let $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. First form the two real paravectors

$$X := a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.1)$$

$$Y := b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.2)$$

Their complex combination is the general complex paravector

$$Z := X + iY = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}. \quad (3.3)$$

Equivalently, collect its complex scalar and vector coefficients as

$$S := a + bi \quad (3.4)$$

$$\mathbf{V} := \mathbf{A} + \mathbf{B}i \quad (3.5)$$

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.6)$$

The corresponding coefficient-extraction operations are

$$\text{sc}(Z) := S \quad (3.7)$$

$$\text{vec}(Z) := \mathbf{V}. \quad (3.8)$$

Sigma conjugation changes the sign of every sigma generator without changing multiplied order; the Hermitian operator changes multiplied sigma order. For the complex coefficients, write

$$S^{\text{H}} = S^* := a - bi \quad (3.9)$$

$$\mathbf{V}^{\text{H}} = \mathbf{V}^* := \mathbf{A} - \mathbf{B}i. \quad (3.10)$$

For a paravector, the two operations are then

$$Z^* := S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (-\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (3.11)$$

$$Z^{\text{H}} := S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (\mathbf{A} - \mathbf{B}i) \cdot \boldsymbol{\sigma}. \quad (3.12)$$

Their composition is the adjugate:

$$\text{adj}(Z) := Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.13)$$

Define the paravector trace and determinant by

$$\text{tr}(Z) := Z + \text{adj}(Z) = 2S, \quad (3.14)$$

$$\det(Z) := \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (3.15)$$

For any paravectors Z and Z' , the adjugate reverses product order:

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z). \quad (3.16)$$

Consequently,

$$\det(ZZ') = \text{adj}(ZZ') ZZ' \quad (3.17)$$

$$= \text{adj}(Z') (\text{adj}(Z) Z) Z' \quad (3.18)$$

$$= \det(Z) \det(Z'). \quad (3.19)$$

For any complex scalar c ,

$$\text{adj}(cZ) = c \text{adj}(Z). \quad (3.20)$$

Therefore

$$\det(cZ) = c^2 \det(Z). \quad (3.21)$$

Whenever $\det(Z) \neq 0$, define

$$Z^{-1} := \frac{\text{adj}(Z)}{\det(Z)} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2} \quad (3.22)$$

$$Z^{-1}Z = ZZ^{-1} = 1. \quad (3.23)$$

A paravector denoted by U is called a unit when it has a two-sided inverse. Its defining property is

$$UU^{-1} = U^{-1}U = 1. \quad (3.24)$$

For a paravector this is equivalent to $\det(U) \neq 0$, with $U^{-1} = \text{adj}(U) / \det(U)$. The definition does not by itself require $\det(U) = 1$ or $U^H = U^{-1}$; either is an additional property of particular units. For another paravector $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, division by Z is generally order-sensitive and must be written using an explicitly ordered inverse. For invertible Z , the unambiguous quotient is defined when Z' and Z commute:

$$\frac{Z'}{Z} := Z'Z^{-1} = Z^{-1}Z' \quad \text{if} \quad \mathbf{V} \times \mathbf{V}' = 0. \quad (3.25)$$

Differentiating $Z^{-1}Z = 1$ gives the general inverse rule

$$d(Z^{-1}) = -Z^{-1} dZ Z^{-1}. \quad (3.26)$$

Both trace and determinant are cyclic:

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (3.27)$$

$$\det(ZZ') = \det(Z'Z). \quad (3.28)$$

For three factors,

$$\text{tr}(ZZ'Z'') = \text{tr}(Z'Z''Z) = \text{tr}(Z''ZZ'), \quad (3.29)$$

$$\det(ZZ'Z'') = \det(Z'Z''Z) = \det(Z''ZZ'). \quad (3.30)$$

For any complex paravector Z and invertible complex paravector T , let

$$Z' := T^{-1}ZT. \quad (3.31)$$

Cyclicity then gives

$$\text{tr}(Z') = \text{tr}(T^{-1}ZT) = \text{tr}(ZTT^{-1}) = \text{tr}(Z), \quad (3.32)$$

$$\det(Z') = \det(T^{-1}ZT) = \det(ZTT^{-1}) = \det(Z). \quad (3.33)$$

Replacing T by T^{-1} gives the same two invariances for $Z' = TZT^{-1}$. Finally, define real and imaginary paravector extraction by

$$\text{re}(Z) := \frac{1}{2}(Z + Z^{\text{H}}) = X, \quad (3.34)$$

$$\text{im}(Z) := \frac{Z - Z^{\text{H}}}{2\text{i}} = Y. \quad (3.35)$$

The functions re and im also act componentwise on complex scalars and vectors. The scalar and vector parts are recovered by

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*\text{H}}), \quad (3.36)$$

$$\mathbf{V} = \text{vec}(Z), \quad (3.37)$$

$$\mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*\text{H}}). \quad (3.38)$$

The four real coefficients then follow by composition:

$$a = \text{re}(S) = \text{sc}(X) = \text{sc}(\text{re}(Z)), \quad (3.39)$$

$$b = \text{im}(S) = \text{sc}(Y) = \text{sc}(\text{im}(Z)), \quad (3.40)$$

$$\mathbf{A} = \text{re}(\mathbf{V}) = \text{vec}(X) = \text{vec}(\text{re}(Z)), \quad (3.41)$$

$$\mathbf{B} = \text{im}(\mathbf{V}) = \text{vec}(Y) = \text{vec}(\text{im}(Z)). \quad (3.42)$$

These definitions are intrinsic to the algebra. Appendix A later reproduces them through the faithful Pauli-matrix representation.

3.2 Products and quadratic forms

Here $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$ are real coefficient vectors. From the ordered sigma multiplication rule,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \quad (3.43)$$

$$= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 A_n B_m \epsilon_{nmk} \text{i} \sigma_k \quad (3.44)$$

$$= \mathbf{A} \cdot \mathbf{B} + \text{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.45)$$

Adding the product with $\mathbf{A}$ and $\mathbf{B}$ interchanged gives

$$2(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}). \quad (3.46)$$

Consequently, for every nonnegative integer n ,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^2 = \mathbf{A}^2, \quad (3.47)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{A}^2)^n, \quad (3.48)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} = (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (3.49)$$

Also,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} - \text{i}\mathbf{A} \times \boldsymbol{\sigma}. \quad (3.50)$$

The corresponding commutator is

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.51)$$

Hence

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] = -2i\mathbf{A} \times \boldsymbol{\sigma}, \quad (3.52)$$

$$[\mathbf{A} \cdot i\boldsymbol{\sigma}, \boldsymbol{\sigma}] = 2\mathbf{A} \times \boldsymbol{\sigma}, \quad (3.53)$$

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.54)$$

3.3 Quadratic identity and eigenvalues

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Since $Z - S = \mathbf{V} \cdot \boldsymbol{\sigma}$ and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$,

$$(Z - S)^2 = \mathbf{V}^2, \quad (3.55)$$

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0. \quad (3.56)$$

Using $\text{tr}(Z) = 2S$ and $\det(Z) = S^2 - \mathbf{V}^2$ gives the equivalent trace-determinant identity

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (3.57)$$

Replacing Z by a scalar eigenvalue λ and solving this quadratic gives

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = S \pm \sqrt{\mathbf{V}^2}. \quad (3.58)$$

Because $\mathbf{V}$ is generally complex, $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0. \quad (3.59)$$

3.4 Real products

Let

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.60)$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.61)$$

Here $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. Then

$$XY = (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.62)$$

$$\text{sc}(XY) = ab + \mathbf{A} \cdot \mathbf{B}, \quad (3.63)$$

$$\text{vec}(XY) = a\mathbf{B} + b\mathbf{A} + i\mathbf{A} \times \mathbf{B}, \quad (3.64)$$

$$[X, Y] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.65)$$

$$X^2 = (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \quad (3.66)$$

$$\text{sc}(X^2) = a^2 + \mathbf{A}^2, \quad (3.67)$$

$$\text{vec}(X^2) = 2a\mathbf{A}, \quad (3.68)$$

$$X^2 - 2aX + (a^2 - \mathbf{A}^2) = 0. \quad (3.69)$$

The condition $\mathbf{A} \times \mathbf{B} = 0$ implies $XY = YX$ and $X^{-1}Y = YX^{-1}$.

3.5 Complex products

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, where $S, S' \in \mathbb{C}$ and $\mathbf{V}, \mathbf{V}' \in \mathbb{C}^3$.

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.70)$$

Thus

$$\text{sc}(ZZ') = SS' + \mathbf{V} \cdot \mathbf{V}', \quad (3.71)$$

$$\text{vec}(ZZ') = S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}'. \quad (3.72)$$

Equivalently, if $Z = X + iY$ and $Z' = X' + iY'$, then

$$ZZ' = (XX' - YY') + i(XY' + YX'). \quad (3.73)$$

Let

$$Z = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (3.74)$$

where $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, and use primes for a second paravector with real primed components. The four real components of ZZ' are

$$\text{real scalar: } aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \quad (3.75)$$

$$\text{imaginary scalar: } ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \quad (3.76)$$

$$\text{real vector: } a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \quad (3.77)$$

$$\text{imaginary vector: } a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \quad (3.78)$$

For the important quadratic products:

$$Z^2 = (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2i(ab + \mathbf{A} \cdot \mathbf{B}) + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.79)$$

$$ZZ^* = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) - 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.80)$$

$$Z^*Z = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) + 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.81)$$

$$ZZ^H = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.82)$$

$$Z^H Z = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.83)$$

$$ZZ^{*H} = \det(Z) = a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (3.84)$$

The scalar part is cyclic, $\text{sc}(ZZ') = \text{sc}(Z'Z)$. The operator laws already established also give $(ZZ')^{*H} = Z'^{*H}Z^{*H}$. The opposite cross-product signs follow the displayed product order. Thus

$ZZ^* = Z^*Z$ and $ZZ^H = Z^H Z$ precisely when $\mathbf{A} \times \mathbf{B} = 0$.

3.5.1 Norms

For $S = a + bi$ and $\mathbf{V} = \mathbf{A} + \mathbf{B}i$, with real components $a, b, \mathbf{A}, \mathbf{B}$, define

$$\|S\|^2 := SS^* = S^*S = a^2 + b^2, \quad (3.85)$$

$$\|\mathbf{V}\|^2 := \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} = \mathbf{A}^2 + \mathbf{B}^2, \quad (3.86)$$

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2. \quad (3.87)$$

Equivalently,

$$\|Z\|^2 = \text{sc}(Z^H Z) = \frac{1}{2} \text{tr}(Z^H Z) = \text{sc}(ZZ^H) = \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*). \quad (3.88)$$

These norms are not algebraic squares; in general,

$$S^2 \neq \|S\|^2 \quad (3.89)$$

$$\mathbf{V}^2 \neq \|\mathbf{V}\|^2. \quad (3.90)$$

For any complex scalar c ,

$$\|cZ\|^2 = \|c\|^2 \|Z\|^2. \quad (3.91)$$

3.5.2 Comparison with the determinant

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$. Its norm is real and nonnegative, whereas $\det(Z)$ may be complex. For $c = re^{i\theta}$ with real r and θ , the determinant retains the phase while the norm does not:

$$\det(re^{i\theta} Z) = r^2 \det(Z) e^{2i\theta}, \quad (3.92)$$

$$\|re^{i\theta} Z\|^2 = r^2 \|Z\|^2. \quad (3.93)$$

3.6 Projection operators

For a real unit vector $\mathbf{u}$, define

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.94)$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\Pi_{\pm}^2 = \Pi_{\pm}, \quad (3.95)$$

$$\Pi_+ \Pi_- = 0, \quad (3.96)$$

$$\Pi_+ + \Pi_- = 1, \quad (3.97)$$

$$\Pi_{\pm}^H = \Pi_{\pm}, \quad (3.98)$$

$$\Pi_{\pm}^* = \Pi_{\mp}, \quad (3.99)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} = \pm \Pi_{\pm}. \quad (3.100)$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. For two unit directions,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.101)$$

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.102)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (3.103)$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. Then

$$\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \quad (3.104)$$

$$\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} = \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \Pi_{\mp}. \quad (3.105)$$

For comparison, define $\Pi'_{\pm} := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})$. Their products are

$$\Pi_+\Pi'_+ = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} + \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.106)$$

$$\Pi_+\Pi'_- = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} - \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.107)$$

$$\Pi_-\Pi'_+ = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} + \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.108)$$

$$\Pi_-\Pi'_- = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} - \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (3.109)$$

For arbitrary real vector increments $d\mathbf{u}$ and $d\mathbf{u}'$, the same dot-cross decomposition gives the geometric determinant identity

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (3.110)$$

Consequently,

$$\sqrt{\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}))} \geq \frac{1}{2} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|. \quad (3.111)$$

This is Lagrange's dot-cross identity written in the algebra; it is not a quantum uncertainty relation.

3.6.1 Ladder operators

For projections along the third axis, define

$$Z_{\pm} = \frac{1}{2}(\sigma_1 \pm i\sigma_2) \quad (3.112)$$

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \sigma_3). \quad (3.113)$$

Then

$$[\sigma_3, Z_{\pm}] = \pm 2Z_{\pm}, \quad (3.114)$$

$$Z_{\pm}^2 = 0, \quad (3.115)$$

$$Z_{\pm}Z_{\mp} = \Pi_{\pm}, \quad (3.116)$$

$$\sigma_1 = Z_+ + Z_-, \quad (3.117)$$

$$\sigma_2 = (Z_+ - Z_-)/i, \quad (3.118)$$

$$\sigma_3 = \Pi_+ - \Pi_-. \quad (3.119)$$

The ladder operators map between the two projected sectors according to

$$Z_+\Pi_+ = 0, \quad (3.120)$$

$$Z_+\Pi_- = Z_+, \quad (3.121)$$

$$\Pi_+Z_+ = Z_+, \quad (3.122)$$

$$\Pi_-Z_+ = 0, \quad (3.123)$$

$$Z_-\Pi_- = 0, \quad (3.124)$$

$$Z_-\Pi_+ = Z_-, \quad (3.125)$$

$$\Pi_-Z_- = Z_-, \quad (3.126)$$

$$\Pi_+Z_- = 0. \quad (3.127)$$

3.7 Functions of paravectors

Here Z denotes the paravector to which the following constructions apply.

3.7.1 General projector rule

Let $\lambda_+, \lambda_- \in \mathbb{C}$. Suppose $\Pi_{\pm}^2 = \Pi_{\pm}$, $\Pi_+ + \Pi_- = 1$, $\Pi_+\Pi_- = \Pi_-\Pi_+ = 0$, and

$$Z = \lambda_+\Pi_+ + \lambda_-\Pi_- . \quad (3.128)$$

Then orthogonality gives

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_- . \quad (3.129)$$

Thus, for a convergent power series $f(w) = \sum_{n=0}^{\infty} c_n w^n$,

$$f(Z) = \sum_{n=0}^{\infty} c_n Z^n \quad (3.130)$$

$$= f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_- . \quad (3.131)$$

3.7.2 Functions of a real unit vector

For a real unit vector $\mathbf{u}$, let $X = \mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. Its eigenvalues are ± 1 . The general rule and $\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$ give

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(1)\Pi_+ + f(-1)\Pi_- = \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma} . \quad (3.132)$$

For the square root, the independent scalar choices $\sqrt{1} = \pm 1$ and $\sqrt{-1} = \pm i$ give

$$\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} = \pm \Pi_+ \pm i \Pi_- . \quad (3.133)$$

3.7.3 Real paravectors

Let $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$, where $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$. For $\mathbf{A} \neq 0$, put

$$\mathbf{u} := \frac{\mathbf{A}}{\|\mathbf{A}\|} \quad (3.134)$$

$$w := \|\mathbf{A}\| . \quad (3.135)$$

Then define

$$\lambda_{\pm} := a \pm w . \quad (3.136)$$

Using $1 = \Pi_+ + \Pi_-$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$ gives

$$X = a(\Pi_+ + \Pi_-) + w(\Pi_+ - \Pi_-) \quad (3.137)$$

$$= \lambda_+\Pi_+ + \lambda_-\Pi_- . \quad (3.138)$$

The general rule therefore gives

$$f(X) = f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_- . \quad (3.139)$$

With nonzero eigenvalues where required and chosen function branches, this includes

$$X^n = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n)\mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.140)$$

$$X^{-1} = \lambda_+^{-1}\Pi_+ + \lambda_-^{-1}\Pi_-, \quad (3.141)$$

$$\sqrt{X} = \pm\sqrt{\lambda_+}\Pi_+ \pm \sqrt{\lambda_-}\Pi_-. \quad (3.142)$$

The exponential and logarithm are developed below. The spectral inverse agrees with the adjugate–determinant formula developed earlier. Indeed,

$$\text{adj}(X) = \lambda_- \Pi_+ + \lambda_+ \Pi_- \quad (3.143)$$

$$\det(X) = \lambda_+ \lambda_-, \quad (3.144)$$

and therefore, when $\lambda_+ \lambda_- \neq 0$,

$$\frac{\text{adj}(X)}{\det(X)} = \lambda_+^{-1}\Pi_+ + \lambda_-^{-1}\Pi_- = X^{-1}. \quad (3.145)$$

For a fixed analytic branch, the function formula reduces continuously to $f(a)$ when $\mathbf{A} = 0$. The projectors themselves are then undefined, and the independent square roots require the separate treatment below.

3.7.4 Direct square-root components

Let

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.146)$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}, \quad (3.147)$$

where $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. Direct multiplication gives

$$X^2 = a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}. \quad (3.148)$$

Thus $X^2 = Y$ requires separate equality of the scalar and vector parts:

$$a^2 + \mathbf{A}^2 = b \quad (3.149)$$

$$2a\mathbf{A} = \mathbf{B}. \quad (3.150)$$

For $\mathbf{B} \neq 0$, the second equation implies $a \neq 0$ and fixes $\mathbf{A}$ to be parallel to $\mathbf{B}$:

$$\mathbf{A} = \frac{\mathbf{B}}{2a}. \quad (3.151)$$

Substitution into the scalar equation gives

$$a^4 - ba^2 + \frac{\mathbf{B}^2}{4} = 0. \quad (3.152)$$

Solving this quadratic in a^2 yields

$$a^2 = \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right). \quad (3.153)$$

For either inner sign, choose either square root of a^2 and then set

$$a = \pm\sqrt{a^2} \quad (3.154)$$

$$\mathbf{A} = \frac{\mathbf{B}}{2a} \quad (3.155)$$

$$\mathbf{A}^2 = b - a^2. \quad (3.156)$$

When $b > \|\mathbf{B}\|$ and $\mathbf{B} \neq 0$, the two choices of a^2 and the two overall signs give four real roots, in agreement with the independent projector signs above. At $b = \|\mathbf{B}\| > 0$, the two values of a^2 coincide and give two distinct real roots. No real root exists when $b < \|\mathbf{B}\|$.

If $\mathbf{B} = 0$, the division by a used above does not cover every possibility. For $b > 0$, the scalar roots $X = \pm\sqrt{b}$ coexist with the pure-vector roots $X = \mathbf{A} \cdot \boldsymbol{\sigma}$ for any $\mathbf{A}^2 = b$. For $b = 0$, the only real root is $X = 0$, and for $b < 0$ there is no real paravector root. The nondegenerate complex case is covered by the spectral construction below; complex-null and scalar cases are stated separately there.

3.7.5 Complex spectral form

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$, $\mathbf{V} \in \mathbb{C}^3$, and $\mathbf{V}^2 \neq 0$, define

$$w := +\sqrt{\mathbf{V}^2}. \quad (3.157)$$

Here the square root is the principal scalar square root, so $w^2 = \mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$. Define

$$\mathbf{u} := \frac{\mathbf{V}}{w}. \quad (3.158)$$

This normalization uses the complex bilinear dot product:

$$\mathbf{u} \cdot \mathbf{u} = \frac{\mathbf{V} \cdot \mathbf{V}}{w^2} = 1. \quad (3.159)$$

Because $\mathbf{u}$ is complex, its Hermitian norm $\|\mathbf{u}\|^2$ need not equal 1. Define

$$\lambda_{\pm} := S \pm w. \quad (3.160)$$

Define

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.161)$$

Because $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = \mathbf{u} \cdot \mathbf{u} = 1$,

$$\Pi_{\pm}^2 = \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = \Pi_{\pm}, \quad (3.162)$$

$$\Pi_+ \Pi_- = \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0. \quad (3.163)$$

Also, $1 = \Pi_+ + \Pi_-$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$, so

$$Z = S(\Pi_+ + \Pi_-) + w(\Pi_+ - \Pi_-) \quad (3.164)$$

$$= \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.165)$$

The general rule gives

$$f(Z) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.166)$$

The adjugate and determinant have the corresponding spectral forms

$$\text{adj}(Z) = \lambda_- \Pi_+ + \lambda_+ \Pi_- \quad (3.167)$$

$$\det(Z) = \lambda_+ \lambda_- . \quad (3.168)$$

Consequently, when $\lambda_+ \lambda_- \neq 0$,

$$\frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_- = Z^{-1}. \quad (3.169)$$

Let $\sqrt{\lambda_+}$ and $\sqrt{\lambda_-}$ denote the principal scalar roots. The two signs below are chosen independently, giving up to four square roots

$$\sqrt{Z} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_- . \quad (3.170)$$

They are four distinct roots when both eigenvalues are nonzero. If one eigenvalue vanishes, its two sign choices coincide.

For completeness, suppose instead that $\mathbf{V}^2 = 0$ and put $N = \mathbf{V} \cdot \boldsymbol{\sigma}$, so $N^2 = 0$. For any scalar function regular at S , the power series terminates in the nilpotent part:

$$f(Z) = f(S) + f'(S)N. \quad (3.171)$$

When $S \neq 0$, this gives

$$\sqrt{Z} = r + \frac{N}{2r}, \quad r^2 = S, \quad (3.172)$$

$$\log Z = \log S + \frac{N}{S}, \quad (3.173)$$

with a consistent scalar branch. If $S = 0$ and $N \neq 0$, the nonzero nilpotent paravector has neither a square root nor a logarithm in this algebra. When $N = 0$, a fixed analytic branch gives the scalar value $f(S)$; additional non-scalar roots of a scalar matrix must be found by solving the square-root equation directly rather than by limiting the projectors.

3.7.6 Exponential of a general complex paravector

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, $w = +\sqrt{\mathbf{V}^2}$, and $\mathbf{u} = \mathbf{V}/w$.

3.7.7 Projection derivation

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, $w = +\sqrt{\mathbf{V}^2}$, $\mathbf{u} = \mathbf{V}/w$, $\lambda_{\pm} = S \pm w$, and $\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The projected function rule (3.166) gives

$$\exp(Z) = \frac{\exp(S+w)}{2}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \frac{\exp(S-w)}{2}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.174)$$

Factoring out $\exp(S)$ and collecting the scalar and vector parts gives

$$\exp(Z) = \exp(S) \left(\frac{\exp(w) + \exp(-w)}{2} + \frac{\exp(w) - \exp(-w)}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.175)$$

3.7.8 Direct series derivation

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, $w = +\sqrt{\mathbf{V}^2}$, and $\mathbf{u} = \mathbf{V}/w$. Because S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$,

$$\exp(Z) = \exp(S) \sum_{n=0}^{\infty} \frac{(w\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!}. \quad (3.176)$$

The identity $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ gives, for every nonnegative integer n ,

$$(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = w^{2n} \quad (3.177)$$

$$(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = w^{2n+1} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.178)$$

Separating the series into even and odd powers therefore gives

$$\exp(Z) = \exp(S) \left(\sum_{n=0}^{\infty} \frac{w^{2n}}{(2n)!} + \left(\sum_{n=0}^{\infty} \frac{w^{2n+1}}{(2n+1)!} \right) \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.179)$$

The scalar definitions of $\cosh w$ and $\sinh w$ reduce both derivations to the same general result:

$$\exp(Z) = \exp(S)(\cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.180)$$

This result does not depend on which root is chosen. Explicitly,

$$\cosh(-w) = \cosh w \quad (3.181)$$

$$\sinh(-w)(-\mathbf{u}) = \sinh w \mathbf{u}. \quad (3.182)$$

Replacing the principal root w by the other algebraic root $-w$ also changes $\mathbf{u} = \mathbf{V}/w$ to $-\mathbf{u}$. Thus both terms in the general result are unchanged.

3.7.9 Component form and real/imaginary limits

For

$$Z = (a + bi) + (\mathbf{A} + \mathbf{Bi}) \cdot \boldsymbol{\sigma}, \quad (3.183)$$

the principal root is

$$w = +\sqrt{\mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}}. \quad (3.184)$$

The corresponding complex bilinear unit direction is

$$\mathbf{u} = \frac{\mathbf{A} + \mathbf{Bi}}{w}. \quad (3.185)$$

The fully expanded formula is

$$\exp(Z) = \exp(a)(\cos b + i \sin b) \left(\cosh w + \frac{\sinh w}{w} (\mathbf{A} + \mathbf{Bi}) \cdot \boldsymbol{\sigma} \right). \quad (3.186)$$

For a real vector, choose $w = \|\mathbf{A}\|$. Then

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \cosh \|\mathbf{A}\| + \sinh \|\mathbf{A}\| \frac{\mathbf{A}}{\|\mathbf{A}\|} \cdot \boldsymbol{\sigma}. \quad (3.187)$$

Because the real scalar a commutes with the vector part, the strictly real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ satisfies

$$\exp(X) = \exp(a) \left(\cosh\|\mathbf{A}\| + \sinh\|\mathbf{A}\| \frac{\mathbf{A}}{\|\mathbf{A}\|} \cdot \boldsymbol{\sigma} \right). \quad (3.188)$$

For an imaginary vector, choose $w = i\|\mathbf{B}\|$. Then

$$\exp(i\mathbf{B} \cdot \boldsymbol{\sigma}) = \cos\|\mathbf{B}\| + i \sin\|\mathbf{B}\| \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.189)$$

Because the imaginary scalar ib also commutes with the vector part, the strictly imaginary paravector $Z = i(b + \mathbf{B} \cdot \boldsymbol{\sigma})$ satisfies

$$\exp(Z) = (\cos b + i \sin b) \left(\cos\|\mathbf{B}\| + i \sin\|\mathbf{B}\| \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma} \right). \quad (3.190)$$

Each quotient in these formulas is understood by its analytic limit at a zero vector.

3.7.10 Kinematic form

Let $\mathbf{A} \in \mathbb{R}^3$ be nonzero. Its real-vector exponential has a direct kinematic interpretation. Define

$$\gamma := \cosh\|\mathbf{A}\| \quad (3.191)$$

$$\mathbf{v} := \tanh\|\mathbf{A}\| \frac{\mathbf{A}}{\|\mathbf{A}\|}. \quad (3.192)$$

Then $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ and

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (3.193)$$

Thus $\|\mathbf{A}\|$ is the rapidity and $\mathbf{A}/\|\mathbf{A}\|$ is the direction of motion. This is the normalized future-directed paravector used below for four-velocity.

3.7.11 Cayley transform

For the real paravector $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, assume $0 < \|\mathbf{B}\| < b$ and put

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|}. \quad (3.194)$$

Because Y is real,

$$Y^* = \text{adj}(Y) = b - \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.195)$$

Its determinant is

$$\det(Y) = Y \text{adj}(Y) = b^2 - \|\mathbf{B}\|^2. \quad (3.196)$$

The assumed range makes $\text{adj}(Y)$ invertible. Its vector part is collinear with that of Y , so the unambiguous quotient notation introduced above applies. Multiplication by Y/Y gives

$$\frac{Y}{\text{adj}(Y)} = \frac{Y}{\text{adj}(Y)} \frac{Y}{Y} = \frac{Y^2}{\det(Y)}. \quad (3.197)$$

The numerator on the right expands as

$$Y^2 = (b + \mathbf{B} \cdot \boldsymbol{\sigma})^2 = b^2 + \|\mathbf{B}\|^2 + 2b\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.198)$$

Dividing this expression by $\det(Y)$ therefore gives

$$\frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} = \frac{b^2 + \|\mathbf{B}\|^2}{b^2 - \|\mathbf{B}\|^2} + \frac{2b\|\mathbf{B}\|}{b^2 - \|\mathbf{B}\|^2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.199)$$

Define

$$\theta := 2 \tanh^{-1} \left(\frac{\|\mathbf{B}\|}{b} \right). \quad (3.200)$$

The double-angle identities give

$$\cosh \theta = \frac{b^2 + \|\mathbf{B}\|^2}{b^2 - \|\mathbf{B}\|^2} \quad (3.201)$$

$$\sinh \theta = \frac{2b\|\mathbf{B}\|}{b^2 - \|\mathbf{B}\|^2}. \quad (3.202)$$

Therefore

$$\begin{aligned} \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\theta) \Pi_+ + \exp(-\theta) \Pi_-. \end{aligned} \quad (3.203)$$

Comparing this result with the kinematic form above identifies its boost-generator vector:

$$\mathbf{A} = \theta \mathbf{u} \quad (3.204)$$

$$\theta = \|\mathbf{A}\|. \quad (3.205)$$

The definition of θ then gives the map from the boost-generator vector $\mathbf{A}$ to the Cayley vector $\mathbf{B}$:

$$\mathbf{B} = b \tanh \left(\frac{\|\mathbf{A}\|}{2} \right) \frac{\mathbf{A}}{\|\mathbf{A}\|}. \quad (3.206)$$

Thus $\mathbf{A}$ is the rapidity vector, whereas $\mathbf{B}/b$ is the Cayley parameter; its magnitude is the hyperbolic tangent of half the rapidity. The choice $b = 1$ recovers the normalized Cayley form, and the case $\mathbf{B} = 0$ follows by continuity.

Finally, the double-angle identities express the kinematic quantities directly in terms of the Cayley variables:

$$\gamma = \frac{b^2 + \|\mathbf{B}\|^2}{b^2 - \|\mathbf{B}\|^2} \quad (3.207)$$

$$\mathbf{v} = \frac{2b\mathbf{B}}{b^2 + \|\mathbf{B}\|^2}. \quad (3.208)$$

Substitution makes the three equivalent forms explicit:

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} = \frac{Y}{Y^*}. \quad (3.209)$$

3.7.12 Aligned boost and rotation

Let $\mathbf{u} \in \mathbb{R}^3$ satisfy $\mathbf{u}^2 = 1$, and let $\phi, \theta \in \mathbb{R}$. Write w as

$$w := \phi + i\theta. \quad (3.210)$$

The parallel real and imaginary generators commute, so

$$\exp(w\mathbf{u} \cdot \boldsymbol{\sigma}) = \exp(\phi\mathbf{u} \cdot \boldsymbol{\sigma}) \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.211)$$

The factors are

$$\exp(\phi\mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh \phi + \sinh \phi \mathbf{u} \cdot \boldsymbol{\sigma} \quad (3.212)$$

$$\exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}) = \cos \theta + i \sin \theta \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.213)$$

Thus ϕ is boost-like and θ is rotation-like. Their product is

$$\exp(w\mathbf{u} \cdot \boldsymbol{\sigma}) = (\cosh \phi \cos \theta + i \sinh \phi \sin \theta) + (\sinh \phi \cos \theta + i \cosh \phi \sin \theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.214)$$

With the conventional factor $1/2$ in the transformation generator, 2ϕ is the rapidity and 2θ is the rotation angle. For complex $\mathbf{u}$, the real and imaginary generator vectors need not be parallel, so this interpretation does not generally apply.

If $\mathbf{V}^2 = 0$, then $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = 0$, even when $\mathbf{V}$ is a nonzero complex-null vector, and the exponential terminates:

$$\exp(Z) = \exp(S)(1 + \mathbf{V} \cdot \boldsymbol{\sigma}). \quad (3.215)$$

Finally, the defining power series respects both conjugation operations:

$$\exp(Z^*) = \exp(Z)^* \quad (3.216)$$

$$\exp(Z^H) = \exp(Z)^H. \quad (3.217)$$

3.7.13 Logarithm of a general complex paravector

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, $w = +\sqrt{\mathbf{V}^2}$, $\mathbf{u} = \mathbf{V}/w$, and $\lambda_{\pm} = S \pm w$. For a nonzero complex scalar λ , each scalar logarithm has the form

$$\log \lambda = \log \|\lambda\| + i(\arg \lambda + 2n\pi), \quad n \in \mathbb{Z}. \quad (3.218)$$

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, $w = +\sqrt{\mathbf{V}^2}$, $\mathbf{u} = \mathbf{V}/w$, and $\lambda_{\pm} = S \pm w$, assume both eigenvalues are nonzero and choose one scalar logarithm of each. The projected function rule gives

$$\log Z = \log(\lambda_+) \Pi_+ + \log(\lambda_-) \Pi_-. \quad (3.219)$$

Substitution of the projectors and collection of the scalar and vector parts gives

$$\log Z = \frac{1}{2} (\log(\lambda_+) + \log(\lambda_-)) + \frac{1}{2} (\log(\lambda_+) - \log(\lambda_-)) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.220)$$

Changing w to $-w$ changes $\mathbf{u}$ to $-\mathbf{u}$ and interchanges λ_+ with λ_- and Π_+ with Π_- , leaving this result unchanged. Independent branch choices for the two scalar logarithms give the different logarithms of Z .

3.7.14 Special cases

Let $\mathbf{u} \in \mathbb{R}^3$ satisfy $\mathbf{u}^2 = 1$, and recall $\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The paravector $\mathbf{u} \cdot \boldsymbol{\sigma}$ has eigenvalues 1 and -1 . Hence all projected branches are

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i \Pi_+ + (2m+1)\pi i \Pi_-, \quad n, m \in \mathbb{Z}. \quad (3.221)$$

For the positive real paravector $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, with $b > \|\mathbf{B}\|$, the real logarithm is

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \tanh^{-1} \left(\frac{\|\mathbf{B}\|}{b} \right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.222)$$

The vector quotient is understood by its limit when $\mathbf{B} = 0$.

The square-root and logarithm formulas above are understood with a fixed analytic branch wherever that branch is regular. Independent branch choices need not have a finite limit when the two eigenvalues coalesce.

4 Transformations and invariants

Here Z is a complex paravector, and T and T' are general complex transformation paravectors. Recall from equation (3.24) that U denotes a unit paravector. A unitary transformation is a unit U that additionally satisfies

$$U^H U = U U^H = 1 \quad (4.1)$$

$$U^H = U^{-1}. \quad (4.2)$$

The transformations used below are organized by the quantities they leave unchanged:

$$\text{tr}(Z) = 2 \text{sc}(Z) \quad (4.3)$$

$$\det(Z) \quad (4.4)$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(Z^H Z). \quad (4.5)$$

The trace is linear; the determinant and norm are quadratic. The prime on T' marks a second transformation factor rather than a derivative. In the physical sections, the same unit notation is specialized locally to proper velocity by recalling $U = dX/ds$.

4.1 Similarity invariants

Throughout this subsection, Z is a complex paravector and the displayed transformation factors act by ordered multiplication.

4.1.1 Trace invariance

Let Z be a complex paravector and T an invertible complex paravector. Consider either similarity:

$$Z' = T^{-1} Z T, \quad (4.6)$$

$$Z' = T Z T^{-1}. \quad (4.7)$$

For example, cyclicity gives

$$\text{tr}(T^{-1} Z T) = \text{tr}(Z T T^{-1}) = \text{tr}(Z). \quad (4.8)$$

Thus $\text{tr}(Z') = \text{tr}(Z)$. For unitary U , the same similarities are

$$Z' = U^H Z U, \quad (4.9)$$

$$Z' = U Z U^H. \quad (4.10)$$

4.1.2 Determinant invariance

Let Z, T, T' be complex paravectors and set $Z' = T' Z T$. Then

$$\det(Z') = \det(T') \det(T) \det(Z). \quad (4.11)$$

Hence $\det(T') \det(T) = 1$ guarantees $\det(Z') = \det(Z)$, and is required if this equality is to hold for arbitrary Z . Equivalently, when $\det(T) \neq 0$, any T' satisfying

$$\det(T') = \frac{1}{\det(T)} \quad (4.12)$$

preserves the determinant of every Z . The automatic choice $T' = T^{-1}$ gives the similarity

$$Z' = T^{-1} Z T. \quad (4.13)$$

The opposite ordering $Z' = T Z T^{-1}$ preserves the determinant as well. For unitary U ,

$$\det(U^H) \det(U) = \det(U^{-1}) \det(U) = 1, \quad (4.14)$$

so both unitary sandwiches preserve the determinant.

4.1.3 Norm invariance

Let Z be a complex paravector and U a unitary paravector satisfying $U^H U = U U^H = 1$. For example, if $Z' = U^H Z U$, then

$$\|Z'\|^2 = \frac{1}{2} \text{tr}(U^H Z^H U U^H Z U) \quad (4.15)$$

$$= \frac{1}{2} \text{tr}(Z^H Z) = \|Z\|^2. \quad (4.16)$$

The same equality $\|Z'\|^2 = \|Z\|^2$ holds for

$$Z' = U Z \quad (4.17)$$

$$Z' = Z U \quad (4.18)$$

$$Z' = U^H Z U \quad (4.19)$$

$$Z' = U Z U^H. \quad (4.20)$$

It does not hold in general when T is nonunitary.

Rotations and boosts preserve different combinations of these invariants. A rotation satisfies

$$R^H = R^{-1} \quad (4.21)$$

$$R^H R = R R^H = 1, \quad (4.22)$$

so R is unitary; $Z' = R^H Z R$ preserves trace, determinant, and norm. A nonzero boost satisfies

$$L^H = L \quad (4.23)$$

$$\det(L) = 1 \quad (4.24)$$

$$L^H L = L^2 \neq 1, \quad (4.25)$$

so L is not unitary. The transformation $Z' = L^{-1} Z L^{-1}$ preserves the determinant but does not generally preserve the trace or norm.

5 Differentiated fields

5.1 Paravector differential operators

In this subsection, $t \in \mathbb{R}$ is a real scalar coordinate and $\mathbf{r} = \{r_1, r_2, r_3\} \in \mathbb{R}^3$ is a real spatial vector. Define their derivatives and the paravector differential operators by

$$\partial_t := \frac{\partial}{\partial t}, \quad \partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}, \quad (5.1)$$

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sum_{n=1}^3 \sigma_n \frac{\partial}{\partial r_n}, \quad (5.2)$$

$$\partial^* = \text{adj}(\partial) = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}. \quad (5.3)$$

We call ∂ the partial paravector and ∂^* its sigma-conjugate partial paravector. They act from the left on every field to their right unless parentheses state otherwise. The sigma generators are constant, while the coefficient derivatives act on the fields.

For fields whose second coordinate derivatives commute, the spatial part squares to a scalar operator:

$$\begin{aligned} (\boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})^2 &= \sum_{n,m=1}^3 \sigma_n \sigma_m \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} + \sum_{n < m} (\sigma_n \sigma_m + \sigma_m \sigma_n) \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} = \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \end{aligned} \quad (5.4)$$

The mixed time-space terms also cancel because the coordinate derivatives commute. Therefore

$$\partial \partial^* = (\partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})(\partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}) = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}, \quad (5.5)$$

$$\partial^* \partial = (\partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})(\partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}) = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \quad (5.6)$$

Thus the two orders give the same scalar second-order operator,

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}} = \frac{\partial^2}{\partial t^2} - \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} =: \square, \quad (5.7)$$

called the d'Alembertian. The equalities here are identities of operators: for example, $\partial \partial^* Z$ means $\partial(\partial^* Z)$.

5.2 Differentiated paravector fields

First let the single paravector field

$$Z(t, \mathbf{r}) = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma} \quad (5.8)$$

be given, with a , b , $\mathbf{A}$, and $\mathbf{B}$ depending on $(t, \mathbf{r})$. Direct use of the ordered vector product gives

$$\partial Z = (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \quad (5.9)$$

$$+ (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \quad (5.10)$$

$$+ i(\partial_{\mathbf{r}} b + \partial_t \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (5.11)$$

Multiplication by i rotates these four components: the real scalar becomes $-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$, the imaginary scalar becomes $\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real vector becomes the negative of the imaginary vector above, and the imaginary vector becomes the real vector.

Now let $Z(t, \mathbf{r})$ and $Z'(t, \mathbf{r})$ be paravector fields; the prime labels the second field and is not a derivative. For their product,

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z', \quad (5.12)$$

$$\partial^*(ZZ') = (\partial^* Z)Z' + Z(\partial^* Z') - [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (5.13)$$

Here the final contraction is defined componentwise, with multiplied order retained:

$$[\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z' := \sum_{n=1}^3 [\sigma_n, Z] \frac{\partial Z'}{\partial r_n}. \quad (5.14)$$

To derive the first rule, apply the ordinary scalar product rule to every coefficient derivative:

$$\begin{aligned} \partial(ZZ') &= (\partial_t Z)Z' + Z(\partial_t Z') \\ &\quad + \sum_{n=1}^3 \sigma_n \left(\frac{\partial Z}{\partial r_n} Z' + Z \frac{\partial Z'}{\partial r_n} \right) \\ &= (\partial Z)Z' + Z(\partial Z') + \sum_{n=1}^3 (\sigma_n Z - Z \sigma_n) \frac{\partial Z'}{\partial r_n} \\ &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \end{aligned} \quad (5.15)$$

The same expansion with the negative spatial part of ∂^* gives

$$\begin{aligned} \partial^*(ZZ') &= (\partial^* Z)Z' + Z(\partial^* Z') \\ &\quad + \sum_{n=1}^3 (-\sigma_n Z + Z \sigma_n) \frac{\partial Z'}{\partial r_n} \\ &= (\partial^* Z)Z' + Z(\partial^* Z') - [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \end{aligned} \quad (5.16)$$

Thus the opposite signs come directly from the opposite spatial signs in ∂ and ∂^* .

As a concrete noncommuting check, take $Z = \sigma_2$ and $Z' = r_1$. Only the r_1 derivative remains, and

$$\sigma_1 \sigma_2 = i\sigma_3, \quad \sigma_2 \sigma_1 = -i\sigma_3, \quad [\sigma_1, \sigma_2] = 2i\sigma_3. \quad (5.17)$$

The first right-hand side gives

$$\sigma_2 \sigma_1 + [\sigma_1, \sigma_2] = -i\sigma_3 + 2i\sigma_3 = i\sigma_3 = \partial(\sigma_2 r_1), \quad (5.18)$$

while the second gives

$$-\sigma_2 \sigma_1 - [\sigma_1, \sigma_2] = i\sigma_3 - 2i\sigma_3 = -i\sigma_3 = \partial^*(\sigma_2 r_1). \quad (5.19)$$

This verifies both the commutator order and the relative sign in a case where Z does not commute with the differential basis.

Thus ∂ and ∂^* obey the ordinary Leibniz rule when the left factor commutes with every σ_n , in particular when it is a complex scalar. For a general paravector field, the commutator term is part of the derivative and multiplied order cannot be rearranged.

6 Space–time kinematics

In this section, $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ is a real spacetime paravector and s is proper time along a time-like worldline.

6.1 Intervals, velocity, and momentum

Represent a differential event by the real paravector

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (6.1)$$

$$dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}. \quad (6.2)$$

The Minkowski interval is

$$ds^2 = \det(dX) = dX \, dX^* = dt^2 - d\mathbf{r}^2. \quad (6.3)$$

For any invertible complex paravector T , define

$$dX' := T^{-1} dX T. \quad (6.4)$$

The cyclic determinant property gives

$$\det(dX') = \det(T^{-1} dX T) = \det(dX T T^{-1}) = \det(dX) = ds^2. \quad (6.5)$$

Thus similarity by any such T leaves the determinant unchanged. This is an algebraic change of basis; even when $T = L$, it is not the physical boost of a space–time paravector. The latter is the congruence $dX' = L^{-1} dX L^{-1}$ developed in Section 9.2. The alternative similarity ordering $T dX T^{-1}$ has the same determinant invariance. For a future-directed time-like interval, choose the positive scalar root

$$ds := \sqrt{\det(dX)}. \quad (6.6)$$

The sign of the invariant squared interval classifies the causal relation between the two neighboring events:

$$ds^2 > 0 : \quad \text{time-like; a massive particle can follow the interval,} \quad (6.7)$$

$$ds^2 = 0 : \quad \text{null; light can follow the interval,} \quad (6.8)$$

$$ds^2 < 0 : \quad \text{space-like; no causal signal can follow the interval.} \quad (6.9)$$

Only in the time-like case is ds real and nonzero under the chosen positive-root convention; along a time-like worldline it is the proper-time differential. Thus the following kinematics explores the determinant of dX and its normalization by this scalar. For two intervals,

$$[dX, dX'] = 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \quad (6.10)$$

When $ds^2 \neq 0$,

$$dX^{-1} = \frac{\text{adj}(dX)}{\det(dX)} = \frac{dX^*}{ds^2}. \quad (6.11)$$

A null interval has $dt^2 = d\mathbf{r}^2$ and no inverse. Along a future-directed time-like worldline, s is proper time. Define the coordinate velocity by

$$\mathbf{v} := \frac{d\mathbf{r}}{dt}. \quad (6.12)$$

Then $d\mathbf{r} = \mathbf{v} dt$, and the positive proper-time increment satisfies

$$ds^2 = (1 - \mathbf{v}^2) dt^2 \quad (6.13)$$

$$ds = \sqrt{1 - \mathbf{v}^2} dt \quad (6.14)$$

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} \quad (6.15)$$

$$ds = \frac{dt}{\gamma} \quad (6.16)$$

$$\gamma = \frac{dt}{ds}. \quad (6.17)$$

For any quantity evaluated along the worldline, the one-variable chain rule therefore gives

$$\frac{d}{ds} = \frac{dt}{ds} \frac{d}{dt} = \gamma \frac{d}{dt}. \quad (6.18)$$

Applying this operator a second time gives

$$\frac{d^2}{ds^2} = \gamma \frac{d}{dt} \left(\gamma \frac{d}{dt} \right) = \gamma^2 \frac{d^2}{dt^2} + \gamma \frac{d\gamma}{dt} \frac{d}{dt}. \quad (6.19)$$

In particular, the proper-velocity paravector is

$$\frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.20)$$

Using the scalar-scaling rule (3.21), its determinant is

$$\det \left(\frac{dX}{ds} \right) = \frac{\det(dX)}{ds^2} = 1. \quad (6.21)$$

Thus dX/ds is a unit paravector in the sense of equation (3.24). We represent this proper velocity by U , so

$$U = \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.22)$$

In terms of γ and $\mathbf{v}$, the same determinant calculation is

$$\det(U) = \det \left(\frac{dX}{ds} \right) = \frac{\det(dX)}{ds^2} = \gamma^2(1 - \mathbf{v}^2) = 1. \quad (6.23)$$

Since U is real, its adjugate is its sigma conjugate, and therefore

$$U^{-1} = \frac{\text{adj}(U)}{\det(U)} = U^* = \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.24)$$

This sigma-conjugate normalization is distinct from Hermitian unitarity; proper velocity is not generally a unitary transformation.

Proper time therefore obeys

$$\tau = \int ds = \int \sqrt{1 - \mathbf{v}^2} dt = \int \frac{dt}{\gamma}. \quad (6.25)$$

For a massive particle with $m > 0$, the four-momentum is the real paravector

$$P := mU = m\gamma + m\gamma\mathbf{v} \cdot \boldsymbol{\sigma} = E + \mathbf{p} \cdot \boldsymbol{\sigma}. \quad (6.26)$$

Its scalar and vector parts define the energy and spatial momentum:

$$E := \text{sc}(P) = m\gamma \quad (6.27)$$

$$\mathbf{p} := \text{vec}(P) = m\gamma\mathbf{v} \quad (6.28)$$

$$\text{adj}(P) = E - \mathbf{p} \cdot \boldsymbol{\sigma}. \quad (6.29)$$

Taking its determinant gives the mass-shell relation

$$\det(P) = m^2 \det(U) = m^2 = \det(E + \mathbf{p} \cdot \boldsymbol{\sigma}) = E^2 - \mathbf{p}^2, \quad (6.30)$$

or $E^2 = \mathbf{p}^2 + m^2$. Since $m > 0$, the momentum is invertible and

$$P^{-1} = \frac{\text{adj}(P)}{\det(P)} = \frac{E - \mathbf{p} \cdot \boldsymbol{\sigma}}{m^2}. \quad (6.31)$$

A massless worldline is null and has no proper-time velocity U . Its four-momentum is therefore defined directly as a future-directed null paravector $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$ satisfying

$$\det(P) = E^2 - \mathbf{p}^2 = 0, \quad E > 0. \quad (6.32)$$

It has no inverse. Thus the same determinant gives the massive and massless dispersion relations, while only the massive momentum is constructed as mU .

6.1.1 Proper acceleration

Recall that $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ is the proper velocity. The required coordinate quantities are

$$\mathbf{v} := \frac{d\mathbf{r}}{dt} \quad (6.33)$$

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} \quad (6.34)$$

$$\gamma := \frac{dt}{ds}. \quad (6.35)$$

Differentiating $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ gives

$$\begin{aligned} \frac{d\gamma}{dt} &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2} \frac{d(1 - \mathbf{v}^2)}{dt} \\ &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2} (-2\mathbf{v} \cdot \mathbf{a}) \\ &= \gamma^3 \mathbf{v} \cdot \mathbf{a}. \end{aligned} \quad (6.36)$$

Hence the second proper-time derivative is

$$\frac{d^2}{ds^2} = \gamma^2 \frac{d^2}{dt^2} + \gamma^4 (\mathbf{v} \cdot \mathbf{a}) \frac{d}{dt}. \quad (6.37)$$

Using $U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ gives the proper-acceleration paravector

$$\begin{aligned} \frac{dU}{ds} &= \frac{d^2 X}{ds^2} = \gamma \frac{d}{dt} (\gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \\ &= \gamma \frac{d\gamma}{dt} (1 + \mathbf{v} \cdot \boldsymbol{\sigma}) + \gamma^2 \mathbf{a} \cdot \boldsymbol{\sigma} \\ &= \gamma^4 (\mathbf{v} \cdot \mathbf{a}) + (\gamma^4 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (6.38)$$

Differentiate the normalization $U^* U = 1$:

$$0 = \frac{d(U^* U)}{ds} = \left(\frac{dU}{ds} \right)^* U + U^* \frac{dU}{ds}. \quad (6.39)$$

Sigma conjugation leaves the scalar part unchanged, and the scalar part is cyclic, so the two scalar terms are equal. Therefore

$$\text{sc}(U^* dU/ds) = 0. \quad (6.40)$$

This is the paravector statement that four-velocity and four-acceleration are Minkowski-orthogonal. Since $U^{-1} = U^*$, right multiplication of the differentiated normalization by U^* gives

$$\text{adj} \left(\frac{dU}{ds} \right) = \left(\frac{dU}{ds} \right)^* = -U^* \frac{dU}{ds} U^*. \quad (6.41)$$

Here adjugation equals sigma conjugation because dU/ds is a real paravector. Direct multiplication gives the scalar-free form

$$U^* \frac{dU}{ds} = \gamma^3 (\mathbf{a} - i \mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}. \quad (6.42)$$

Its real vector contains the coordinate acceleration, while its imaginary vector records the turning of the velocity direction. Since $\mathbf{a} \cdot (\mathbf{v} \times \mathbf{a}) = 0$, its square is scalar. The adjugate identity therefore yields the determinant without a separate component expansion:

$$\begin{aligned} \det \left(\frac{dU}{ds} \right) &= \text{adj} \left(\frac{dU}{ds} \right) \frac{dU}{ds} \\ &= - \left(U^* \frac{dU}{ds} \right)^2 \\ &= -\gamma^6 (\mathbf{a}^2 - (\mathbf{v} \times \mathbf{a})^2) \\ &= -\gamma^4 (\mathbf{a}^2 + \gamma^2 (\mathbf{v} \cdot \mathbf{a})^2). \end{aligned} \quad (6.43)$$

The last step uses Lagrange's cross-product identity $(\mathbf{v} \times \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}^2 - (\mathbf{v} \cdot \mathbf{a})^2$, where each vector square denotes its Euclidean dot product with itself. Geometrically, this follows from $\|\mathbf{v} \times \mathbf{a}\| = \|\mathbf{v}\| \|\mathbf{a}\| \sin \theta$ and $\mathbf{v} \cdot \mathbf{a} = \|\mathbf{v}\| \|\mathbf{a}\| \cos \theta$. Together with $1 - \mathbf{v}^2 = \gamma^{-2}$, it shows that the proper acceleration is space-like unless it vanishes and that $-\det(dU/ds)$ is its invariant squared magnitude.

For $\mathbf{v} \neq 0$, resolve the coordinate acceleration parallel and perpendicular to the velocity by

$$\mathbf{u} := \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad (6.44)$$

$$\mathbf{a}_{\parallel} := (\mathbf{a} \cdot \mathbf{u}) \mathbf{u} \quad (6.45)$$

$$\mathbf{a}_{\perp} := \mathbf{a} - \mathbf{a}_{\parallel}. \quad (6.46)$$

Since $(\mathbf{v} \cdot \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}_{\parallel}^2$ and $1 + \gamma^2 \mathbf{v}^2 = \gamma^2$,

$$\det \left(\frac{dU}{ds} \right) = -\gamma^4 \left(\mathbf{a}_{\perp}^2 + \gamma^2 \mathbf{a}_{\parallel}^2 \right). \quad (6.47)$$

At $\mathbf{v} = 0$, this reduces continuously to $\det(dU/ds) = -\mathbf{a}^2$.

6.2 Varying transformations and second differentials

Recall $U = dX/ds$, and likewise write $U' = dX'/ds$. Let X be a paravector along a path and T an invertible transformation paravector that may vary along it. Let d denote successive differentiation with respect to the path parameter and define

$$dX' = T^{-1} dX T. \quad (6.48)$$

Although $\det(dX') = \det(dX)$ at each point, a varying T adds a term to the second differential. Define

$$d\Theta := T^{-1} dT \quad (6.49)$$

$$dT^{-1} = -d\Theta T^{-1}. \quad (6.50)$$

Direct differentiation gives

$$d^2 X' = d(T^{-1} dX T) \quad (6.51)$$

$$= T^{-1} d^2 X T - d\Theta dX' + dX' d\Theta \quad (6.52)$$

$$= T^{-1} d^2 X T + [dX', d\Theta]. \quad (6.53)$$

Dividing by the corresponding powers of ds , define

$$\Omega := \frac{d\Theta}{ds} = T^{-1} \frac{dT}{ds}, \quad (6.54)$$

so

$$\frac{dU'}{ds} = T^{-1} \frac{dU}{ds} T + [U', \Omega]. \quad (6.55)$$

For constant T , the connection term vanishes. For varying T , it is the algebraic source of the additional terms in rotating and accelerating frames and is the pathwise prototype of a covariant derivative.

Here dX' is defined by the pointwise transformation of the tangent. When T varies, it is not generally $d(T^{-1} X T)$, which also contains the transformation of X itself. Moreover, a general complex T can make dX' complex even when dX is real, so a physical space–time transformation requires an additional reality-preserving condition.

The corresponding local algebra basis is

$$\sigma'_n := T^{-1} \sigma_n T. \quad (6.56)$$

Similarity preserves all products, so the transformed generators obey the same sigma algebra. If T varies along the path, direct differentiation gives

$$d\sigma'_n = [\sigma'_n, d\Theta] \quad (6.57)$$

$$d\Theta = T^{-1} dT. \quad (6.58)$$

This equation describes the change of a local algebra basis. It should not be identified with a physical Lorentz transformation until the action of T and the condition preserving real space–time paravectors have both been specified. In particular, the complex boost-like terms obtained from a completely general T are connection coefficients, not automatically components of a real physical velocity.

6.3 Free-particle action

Let m be the particle mass, $\mathbf{v} = d\mathbf{r}/dt$, $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, and $ds = dt/\gamma$. The free-particle action and Lagrangian are

$$\int L dt = -m \int ds \quad (6.59)$$

$$L = -m\sqrt{1 - \mathbf{v}^2} = -\frac{m}{\gamma}. \quad (6.60)$$

For a general Lagrangian $L(t, \mathbf{r}, \mathbf{v})$,

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L \quad (6.61)$$

$$H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \quad (6.62)$$

$$\frac{dH}{dt} = -\partial_t L. \quad (6.63)$$

Thus spatial translation invariance conserves canonical momentum and time translation invariance conserves the Hamiltonian. For the free-particle Lagrangian,

$$\partial_{\mathbf{v}} L = m\gamma \mathbf{v}, \quad (6.64)$$

$$H = m\gamma \mathbf{v}^2 + \frac{m}{\gamma} = m\gamma(\mathbf{v}^2 + \gamma^{-2}) = m\gamma. \quad (6.65)$$

At low velocity, the binomial expansions are

$$\sqrt{1 - \mathbf{v}^2} = 1 - \frac{1}{2}\mathbf{v}^2 + O(v^4) \quad (6.66)$$

$$\gamma = 1 + \frac{1}{2}\mathbf{v}^2 + O(v^4). \quad (6.67)$$

Multiplying the first expansion by $-m$ gives

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m + \frac{1}{2}m\mathbf{v}^2 + O(v^4), \quad (6.68)$$

while the second gives

$$H = m\gamma = m + \frac{1}{2}m\mathbf{v}^2 + O(v^4). \quad (6.69)$$

The exact relativistic kinetic energy is therefore

$$K := H - m = m(\gamma - 1), \quad (6.70)$$

and, apart from the constant rest-energy term, both expansions recover the nonrelativistic kinetic energy $\frac{1}{2}m\mathbf{v}^2$.

7 Electromagnetism

Throughout this section, $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is the field, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ is the potential, and $D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$ is the source. For the particle equations, recall $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ from equation (6.22).

7.1 Field equation and potentials

Recall the paravector differential operator from equation (5.3), and define the electromagnetic field and source by

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \quad (7.1)$$

$$F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma} \quad (7.2)$$

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (7.3)$$

All four Maxwell equations are the single equation

$$\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (7.4)$$

Sigma conjugation gives the equivalent equation

$$\partial^* F^* = D. \quad (7.5)$$

Expanding the left side by component gives

$$\partial F = \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i\partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \quad (7.6)$$

$$+ \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (7.7)$$

$$+ \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (7.8)$$

Equating these four components with those of $\rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives, respectively,

$$\text{real scalar: } \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (7.9)$$

$$\text{imaginary scalar: } \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (7.10)$$

$$\text{real vector: } \partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad (7.11)$$

$$\text{imaginary vector: } \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (7.12)$$

Equation (5.7) gives $\partial\partial^* = \partial^*\partial = \square$, whereas

$$\partial^2 = \partial_t^2 + \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}} + 2\partial_t \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \neq \square. \quad (7.13)$$

Applying ∂^* to (7.4) gives

$$\square F = \partial^* D^* \quad (7.14)$$

$$= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i\partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}. \quad (7.15)$$

The real-scalar component is charge conservation, and the real- and imaginary-vector components are the electric and magnetic wave equations:

$$\text{real scalar: } \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (7.16)$$

$$\text{real vector: } \square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (7.17)$$

$$\text{imaginary vector: } \square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (7.18)$$

In vacuum, $\rho = 0$ and $\mathbf{J} = 0$, so

$$\square \mathbf{E} = 0 \quad (7.19)$$

$$\square \mathbf{B} = 0. \quad (7.20)$$

For the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$,

$$\partial \Phi = \underbrace{\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}}_{\text{real scalar}} + \underbrace{(\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (7.21)$$

$$+ \underbrace{i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (7.22)$$

Define the scalar component

$$S := \text{sc}(\partial \Phi) = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (7.23)$$

Then $\partial \Phi = S + F^*$: its real-vector and imaginary-vector components give

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V \quad (7.24)$$

$$\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (7.25)$$

Also, $F = \partial^* \Phi^* - S$, so

$$\partial F = \partial \partial^* \Phi^* - \partial S = \square \Phi^* - \partial S \quad (7.26)$$

$$= (\square V - \partial_t S) - (\square \mathbf{A} + \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}. \quad (7.27)$$

Comparing its real-scalar and real-vector components with $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives

$$\rho = \square V - \partial_t S \quad (7.28)$$

$$\mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (7.29)$$

For a scalar field λ , the gauge transformation is

$$\Phi' = \Phi + \partial^* \lambda, \quad (7.30)$$

$$\partial \Phi' = \partial \Phi + \partial \partial^* \lambda = \partial \Phi + \square \lambda. \quad (7.31)$$

Because $\square \lambda$ is scalar, only the scalar component changes:

$$S' = S + \square \lambda \quad (7.32)$$

$$\text{vec}(\partial \Phi') = \text{vec}(\partial \Phi). \quad (7.33)$$

Thus F is unchanged. In the Lorenz gauge $S = 0$, equation (7.28) reduces to

$$\rho = \square V \quad (7.34)$$

$$\mathbf{J} = \square \mathbf{A}. \quad (7.35)$$

7.2 Field invariants

Recall $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Electric and magnetic fields mix under Lorentz transformations, so their individual magnitudes depend on the observer. Two combinations remain the same for every inertial observer and therefore classify the electromagnetic field itself.

The scalar part of F vanishes, and therefore $\text{adj}(F) = -F$. Hence

$$\det(F) = -F^2 \quad (7.36)$$

$$= -(\mathbf{E} + i\mathbf{B}) \cdot (\mathbf{E} + i\mathbf{B}) \quad (7.37)$$

$$= -\mathbf{E}^2 + \mathbf{B}^2 - 2i\mathbf{E} \cdot \mathbf{B}. \quad (7.38)$$

For any invertible real or complex paravector T , let

$$F' = T^{-1}FT. \quad (7.39)$$

The general similarity identity (3.33) gives

$$\det(F') = \det(T^{-1}FT) = \det(F). \quad (7.40)$$

Rotations and boosts are the physical special cases

$$F' = R^{-1}FR = R^HFR \quad (7.41)$$

$$F' = L^{-1}FL = L^*FL, \quad (7.42)$$

because $R^{-1} = R^H$ and $L^{-1} = L^*$. Hence every inertial observer agrees on

$$a = \mathbf{E}^2 - \mathbf{B}^2 = -\text{re}(\det(F)), \quad (7.43)$$

$$b = \mathbf{E} \cdot \mathbf{B} = -\frac{1}{2}\text{im}(\det(F)). \quad (7.44)$$

For nonzero fields, these invariants give the standard classification:

$$b = 0 : \quad \text{electric-like when } a \text{ is positive; some frame has no magnetic field,} \quad (7.45)$$

$$b = 0 : \quad \text{magnetic-like when } a \text{ is negative; some frame has no electric field,} \quad (7.46)$$

$$b \neq 0 : \quad \text{no frame can eliminate either field.} \quad (7.47)$$

When both a and b vanish, the field is null or radiation-like: the electric and magnetic fields are orthogonal and have equal magnitude. Thus the invariants reveal physical properties that cannot be inferred from one observer's values of $\mathbf{E}$ and $\mathbf{B}$ alone. Unlike the energy density and flux below, they do not change with the observer.

7.3 Lorentz force from the field and action

Recall $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$, $P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, and $\mathbf{p} = m\gamma\mathbf{v}$, where s is proper time, $m > 0$ is constant, and q is the particle charge. The Lorentz proper-force equation is

$$\frac{dP}{ds} = m \frac{dU}{ds} = q \text{re}(FU) = \frac{q}{2}(FU + UF^H). \quad (7.48)$$

The Hermitian extraction removes the imaginary magnetic-dual components. The remaining scalar and vector components are, respectively,

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot \mathbf{E} \quad (7.49)$$

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.50)$$

The same result follows from the potential through the interacting-particle action

$$\int L dt = -m \int ds - q \int \text{sc}(\Phi^* U) ds \quad (7.51)$$

$$= \int \left(-\frac{m}{\gamma} - q \text{sc}(\Phi^* U / \gamma) \right) dt \quad (7.52)$$

$$= \int \left(-\frac{m}{\gamma} - q \text{sc}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right) dt \quad (7.53)$$

$$= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt. \quad (7.54)$$

Thus

$$L = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}) \quad (7.55)$$

$$\partial_v L = m\gamma\mathbf{v} + q\mathbf{A} = \mathbf{p} + q\mathbf{A}. \quad (7.56)$$

The Euler–Lagrange terms are

$$\frac{d}{dt} \partial_v L = \frac{d\mathbf{p}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_r) \mathbf{A}, \quad (7.57)$$

$$\partial_r L = -q\partial_r V + q\partial_r(\mathbf{v} \cdot \mathbf{A}). \quad (7.58)$$

Using $(\mathbf{v} \cdot \partial_r) \mathbf{A} = \partial_r(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_r \times \mathbf{A})$ gives

$$\frac{d\mathbf{p}}{dt} = -q\partial_r V - q\partial_t \mathbf{A} + q(\partial_r(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_r) \mathbf{A}) \quad (7.59)$$

$$= q(-\partial_r V - \partial_t \mathbf{A} + \mathbf{v} \times (\partial_r \times \mathbf{A})) \quad (7.60)$$

$$= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.61)$$

The Hamiltonian follows with the velocity-dependent vector-potential terms cancelling:

$$H = \mathbf{v} \cdot \partial_v L - L \quad (7.62)$$

$$= m\gamma\mathbf{v}^2 + q\mathbf{v} \cdot \mathbf{A} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \quad (7.63)$$

$$= m\gamma(\mathbf{v}^2 + \gamma^{-2}) + qV = m\gamma + qV. \quad (7.64)$$

This is the canonical energy and is gauge dependent under time-dependent gauge changes; the kinetic energy $E = m\gamma$ is gauge invariant. A conserved canonical energy is obtained when the external potentials are time independent in the chosen gauge. Finally, $dH/dt = -\partial_t L$ gives

$$\frac{d(m\gamma)}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_r V = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \quad (7.65)$$

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot (-\partial_r V - \partial_t \mathbf{A}) = q\mathbf{v} \cdot \mathbf{E}. \quad (7.66)$$

To see equation (7.48) componentwise, expand

$$\frac{FU}{\gamma} = \mathbf{v} \cdot \mathbf{E} + i\mathbf{v} \cdot \mathbf{B} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + i(\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (7.67)$$

Taking the real paravector part removes the two imaginary components.

7.4 Proper acceleration from the Lorentz force

For a particle of constant mass $m > 0$, recall

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \quad (7.68)$$

$$U = \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad (7.69)$$

$$P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}, \quad (7.70)$$

$$E = m\gamma, \quad \mathbf{p} = m\gamma\mathbf{v}, \quad (7.71)$$

$$\frac{dP}{ds} = m \frac{dU}{ds}. \quad (7.72)$$

Also let $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ and $\mathbf{a} = d\mathbf{v}/dt$. Thus dP/ds is the proper-force paravector, while dU/ds is the proper-acceleration paravector.

The Lorentz four-force is automatically orthogonal to the four-velocity. Since $U^*U = UU^* = 1$, cyclicity of the scalar part and $\text{sc}(F) = \text{sc}(F^H) = 0$ give

$$\text{sc}(U^* dP/ds) = q \text{sc}(U^* \text{re}(FU)) \quad (7.73)$$

$$= \frac{q}{2} \text{sc}(U^*FU + U^*UF^H) \quad (7.74)$$

$$= \frac{q}{2} (\text{sc}(F) + \text{sc}(F^H)) = 0. \quad (7.75)$$

Because $dP/ds = m dU/ds$, this is equivalently

$$\text{sc}(U^* dP/ds) = m \text{sc}(U^* dU/ds) = 0, \quad (7.76)$$

in agreement with the proper-acceleration development above.

The earlier kinematic result for dU/ds , multiplied by m , gives

$$\frac{dP}{ds} = \gamma \left(\frac{dE}{dt} + \frac{d\mathbf{p}}{dt} \cdot \boldsymbol{\sigma} \right) \quad (7.77)$$

$$= m\gamma^4(\mathbf{v} \cdot \mathbf{a}) + m(\gamma^2\mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (7.78)$$

The real paravector part of the field product is

$$q \text{re}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}). \quad (7.79)$$

Equating the scalar and vector parts of $dP/ds = q \text{re}(FU)$ and dividing by γ gives

$$\frac{dE}{dt} = m\gamma^3\mathbf{v} \cdot \mathbf{a} = q\mathbf{v} \cdot \mathbf{E}, \quad (7.80)$$

$$\frac{d\mathbf{p}}{dt} = m\gamma(\mathbf{a} + \gamma^2(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.81)$$

Eliminating $\mathbf{v} \cdot \mathbf{a}$ between these two equations yields

$$\mathbf{a} = \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v}). \quad (7.82)$$

The magnetic term is perpendicular to $\mathbf{v}$ and therefore changes momentum without contributing to the coordinate-time power $dE/dt = q\mathbf{v} \cdot \mathbf{E}$.

The scalar and vector parts of the proper force give its invariant squared magnitude directly:

$$-\det\left(\frac{dP}{ds}\right) = q^2\gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2). \quad (7.83)$$

Since $dP/ds = m dU/ds$, the scalar-scaling rule gives

$$\det\left(\frac{dP}{ds}\right) = m^2 \det\left(\frac{dU}{ds}\right). \quad (7.84)$$

Therefore the corresponding invariant squared proper-acceleration magnitude is

$$-\det\left(\frac{dU}{ds}\right) = \frac{q^2}{m^2}\gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2). \quad (7.85)$$

The factor $\gamma^2((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2)$ is the squared electric-field magnitude in the particle's instantaneous rest frame. Thus dP/ds is the space-like proper force required by the fixed mass shell $P^*P = m^2$; division by m gives the proper acceleration required by $U^*U = 1$.

7.5 Energy balance

Since

$$F = (\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma} \quad (7.86)$$

$$F^H = (\mathbf{E} - \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, \quad (7.87)$$

their ordered product is

$$FF^H = \mathbf{E}^2 + \mathbf{B}^2 + 2(\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (7.88)$$

Therefore the standard electromagnetic energy density u and Poynting vector $\mathbf{S}$ are the scalar and vector coefficients of $\frac{1}{2}FF^H$:

$$u := \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) = \frac{1}{2}\text{sc}(FF^H), \quad (7.89)$$

$$\mathbf{S} := \mathbf{E} \times \mathbf{B} = \frac{1}{2}\text{vec}(FF^H). \quad (7.90)$$

These quantities describe energy measured in the chosen frame; they are not the two Lorentz invariants contained in F^2 . The product $\frac{1}{2}FF^H$ packages energy density and energy flux into a paravector, but is not by itself the full electromagnetic stress-energy tensor. The full tensor is represented without indices by the linear map

$$T(N) := \frac{1}{2}FNF^H, \quad (7.91)$$

where N is a real paravector. In the frame whose time direction is $N = 1$, $T(1) = u + \mathbf{S} \cdot \boldsymbol{\sigma}$. If $F' = L^{-1}FL$ and $N' = L^{-1}NL^{-1}$, then

$$T'(N') = L^{-1}T(N)L^{-1}, \quad (7.92)$$

so every energy–momentum flux paravector has the correct four-vector law. Its time-direction value is future causal because

$$\det(T(1)) = u^2 - \mathbf{S}^2 = \frac{1}{4} \|\det(F)\|^2 \geq 0. \quad (7.93)$$

Equality holds for a null radiation field. Thus the field invariants, energy density, and Poynting flux meet in one determinant identity. This product order gives the conventional sign of the Poynting vector directly. For comparison, reversing the order preserves the scalar coefficient and changes the sign of the vector coefficient:

$$\text{sc}(F^H F) = \text{sc}(F F^H) \quad (7.94)$$

$$\text{vec}(F^H F) = -\text{vec}(F F^H). \quad (7.95)$$

Starting only from the paravector field equation $\partial F = D^*$, its real-vector and imaginary-vector components give

$$\partial_t \mathbf{E} = \partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J} \quad (7.96)$$

$$\partial_t \mathbf{B} = -\partial_{\mathbf{r}} \times \mathbf{E}. \quad (7.97)$$

Taking their dot products with $\mathbf{E}$ and $\mathbf{B}$, respectively, yields

$$\partial_t u = \mathbf{E} \cdot \partial_t \mathbf{E} + \mathbf{B} \cdot \partial_t \mathbf{B} \quad (7.98)$$

$$= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B}) - \mathbf{E} \cdot \mathbf{J} - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \quad (7.99)$$

$$= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E}, \quad (7.100)$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Substitution into $\partial_t u$ therefore derives Poynting's theorem, the local electromagnetic energy-balance law:

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.101)$$

Here $\mathbf{J} \cdot \mathbf{E}$ is the power density transferred from the field to matter; the minus sign is the corresponding loss of field energy. Equivalently, in paravector coefficients,

$$\frac{1}{2} \partial_t \text{sc}(F F^H) + \frac{1}{2} \partial_{\mathbf{r}} \cdot \text{vec}(F F^H) = -\mathbf{J} \cdot \mathbf{E}. \quad (7.102)$$

8 Relativistic wave equations

Recall that $U = dX/ds$ for a classical massive particle. For a free massive future mass-shell momentum, the corresponding normalized unit is $U = P/m$.

In this section, ψ denotes a scalar or paravector amplitude as stated locally, Ψ denotes a two-paravector Dirac block, m is the mass, and q is the charge. Three uses of the algebra must be kept distinct. Real space–time quantities such as X , P , and a conserved current transform as four-vectors. Scalar and pure-vector fields carry spin zero and spin one through their field transformation laws. A spin-one-half state is instead a minimal-ideal amplitude with a one-sided rotation law; a Dirac state contains two such amplitudes with opposite boost laws. The algebra element alone does not determine the spin—its transformation law does.

The wave equations below describe classical fields and, where the conserved density is positive, one-particle wave mechanics in a prescribed electromagnetic field. The negative-frequency solutions required by the relativistic equations are also displayed. Their interpretation as positive-energy antiparticles is stated separately because that step belongs to charge conjugation and field quantization, not to a negative probability.

8.1 Paravector inner products and expectations

For finite paravector amplitudes ψ, ψ' and an algebraic paravector Z , define the local pairing and matrix element intrinsically by scalar extraction:

$$\langle \psi' | \psi \rangle := \text{sc}(\psi'^H \psi), \quad (8.1)$$

$$\langle \psi' | Z | \psi \rangle := \text{sc}(\psi'^H Z \psi). \quad (8.2)$$

The first is the $Z = 1$ case of the second, and $\langle \psi | \psi \rangle = \|\psi\|^2$. In particular,

$$n := \text{sc}(\psi^H \psi) \geq 0 \quad (8.3)$$

is the local norm density. A finite spin amplitude is normalized by $n = 1$, not by $\psi^H \psi = 1$. For spatial wavefunctions the same pairing means

$$\langle \psi' | \psi \rangle := \int \text{sc}(\psi'^H \psi) \, d^3 \mathbf{r}, \quad (8.4)$$

$$\langle \psi' | Z | \psi \rangle := \int \text{sc}(\psi'^H Z \psi) \, d^3 \mathbf{r}. \quad (8.5)$$

Thus wavefunction normalization is $\int n \, d^3 \mathbf{r} = 1$ whenever this positive pairing is the conserved norm, as it is for Pauli and Dirac amplitudes. The Klein–Gordon scalar has a different conserved current and is treated separately below. For an observable $Z^H = Z$ acting from the left,

$$\langle Z \rangle_\psi := \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle}. \quad (8.6)$$

For a finite amplitude this is the local ratio

$$\langle Z \rangle_\psi = \frac{\text{sc}(\psi^H Z \psi)}{\text{sc}(\psi^H \psi)}, \quad (8.7)$$

whereas for a spatial wavefunction both scalar pairings are integrated. For normalized ψ , this reduces to $\langle Z \rangle_\psi = \langle \psi | Z | \psi \rangle$. If an observable contains derivatives, those derivatives act on the amplitude to their right. Hermiticity is then a statement about the integrated pairing and follows by integration by parts with suitable boundary conditions; it is not obtained by applying H only to the displayed algebra coefficient. For unrestricted paravectors, the local pairing is one half of the usual Hilbert–Schmidt pairing of their matrix images. A single Pauli spinor is obtained only after restricting the amplitude to a minimal left ideal as described next. Appendix A.4 then shows that these definitions become the standard two-component column formulas.

For a paravector T with $\psi' = T\psi$, a finite amplitude obeys

$$\langle Z \rangle_{\psi'} = \frac{\text{sc}(\psi^H T^H Z T \psi)}{\text{sc}(\psi^H T^H T \psi)}. \quad (8.8)$$

For a spatial wavefunction, integrate the scalar numerator and denominator before taking their ratio. If $T^H T = 1$ and $\langle \psi | \psi \rangle = 1$, then the denominator is unity and

$$\langle Z \rangle_{\psi'} = \text{sc}(\psi^H T^H Z T \psi). \quad (8.9)$$

Thus a unitary rotation preserves the normalization. A finite boost is not unitary, so this positive equal-time norm is not itself Lorentz invariant; the covariant Dirac normalization instead comes from the conserved Dirac current on a spacelike hypersurface.

8.2 Physical spinors and minimal ideals

In this subsection, Π_3 is the fixed column idempotent

$$\Pi_3 := \frac{1}{2}(1 + \sigma_3). \quad (8.10)$$

It satisfies $\Pi_3^2 = \Pi_3 = \Pi_3^H$ and is held fixed; it selects a column representation when it multiplies from the right. When the same element acts from the left it is also the positive- σ_3 measurement projector; the side of multiplication distinguishes these two roles. A physical two-component Pauli amplitude has the form

$$\psi = Z\Pi_3, \quad \psi\Pi_3 = \psi, \quad (8.11)$$

where $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$, is an unrestricted complex paravector, not a scalar. The representative Z is not unique; the product $Z\Pi_3$ contains the two complex components of ψ . Let $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, with $S' \in \mathbb{C}$ and $\mathbf{V}' \in \mathbb{C}^3$, be any unrestricted complex paravector. These amplitudes form a minimal left ideal because every such left action remains in the same set:

$$(Z'\psi)\Pi_3 = Z'(\psi\Pi_3) = Z'\psi. \quad (8.12)$$

Right multiplication by the fixed Π_3 is therefore essential. Left multiplication by Π_3 would instead produce a right ideal, which is not preserved by general observables acting from the left.

A normalized basis of this ideal is

$$e_+ := \sqrt{2}\Pi_3, \quad e_- := \sqrt{2}\sigma_1\Pi_3. \quad (8.13)$$

Direct use of the algebra operations gives

$$\text{sc}(e_+^H e_+) = \text{sc}(e_-^H e_-) = 1, \quad (8.14)$$

$$\text{sc}(e_+^H e_-) = 0, \quad (8.15)$$

$$\sigma_3 e_+ = e_+, \quad \sigma_3 e_- = -e_-. \quad (8.16)$$

Every ideal amplitude can therefore be written with two complex scalars as

$$\psi = c_+ e_+ + c_- e_-. \quad (8.17)$$

For a σ_3 measurement its two probabilities are simply

$$p_+ = \frac{\|c_+\|^2}{\|c_+\|^2 + \|c_-\|^2}, \quad p_- = \frac{\|c_-\|^2}{\|c_+\|^2 + \|c_-\|^2}. \quad (8.18)$$

Appendix A.4 maps e_+ and e_- to the standard columns $\{1, 0\}^\top$ and $\{0, 1\}^\top$.

For a physical spin measurement, let $\mathbf{u} \in \mathbb{R}^3$ be a unit direction and use the projectors from Section 3.6,

$$\Pi_\pm := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (8.19)$$

They satisfy the identities listed there, including $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+ \Pi_- = 0$, and $\Pi_\pm^H = \Pi_\pm$. They act from the left and give

$$\psi_\pm := \Pi_\pm \psi. \quad (8.20)$$

The algebra element $\mathbf{u} \cdot \boldsymbol{\sigma}$ has eigenvalues $+1$ and -1 . In natural units the physical spin component is one half of this element, so spin up and spin down mean respectively

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi = \psi, \quad \Pi_+ \psi = \psi, \quad \Pi_- \psi = 0, \quad (8.21)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi = -\psi, \quad \Pi_- \psi = \psi, \quad \Pi_+ \psi = 0. \quad (8.22)$$

The corresponding measured spin values are $+\frac{1}{2}$ and $-\frac{1}{2}$. Because $\psi_{\pm}\Pi_3 = \psi_{\pm}$, both outcomes remain in the same fixed minimal left ideal. Hermitian conjugation reverses the multiplied order, so their norms and observable matrix elements are

$$\langle \psi_{\pm} | \psi_{\pm} \rangle = \text{sc}(\psi^{\text{H}}\Pi_{\pm}^{\text{H}}\Pi_{\pm}\psi) = \langle \psi | \Pi_{\pm} | \psi \rangle, \quad (8.23)$$

$$\langle \psi_{\pm} | Z' | \psi_{\pm} \rangle = \text{sc}(\psi^{\text{H}}\Pi_{\pm}^{\text{H}}Z'\Pi_{\pm}\psi) = \langle \psi | \Pi_{\pm}Z'\Pi_{\pm} | \psi \rangle. \quad (8.24)$$

The spin direction of a finite amplitude also has a compact intrinsic form. For nonzero ψ , define the normalized Hermitian paravector

$$B := \frac{\psi\psi^{\text{H}}}{\text{sc}(\psi^{\text{H}}\psi)} = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}, \quad \mathbf{s} := \text{vec}(B). \quad (8.25)$$

It obeys $B^{\text{H}} = B$ and $\text{sc}(B) = 1$. Multiplicativity gives the entirely intrinsic identity

$$\det(B) = \frac{\det(\psi)\det(\psi^{\text{H}})}{n^2} = \frac{\|\det(\psi)\|^2}{n^2} = 1 - \mathbf{s}^2 \geq 0. \quad (8.26)$$

A nonzero minimal-ideal amplitude has $\det(\psi) = 0$, so $\mathbf{s}^2 = 1$ and $\mathbf{s}$ is the unit Bloch direction of a pure Pauli state. For an unrestricted amplitude, $\mathbf{s}^2 \leq 1$ and B can also represent a mixed state. Thus pure states lie on the null boundary $\det(B) = 0$, mixed states lie inside it, and the Bloch ball is the unit-scalar slice of this paravector cone. This is state geometry, not the particle's spacetime momentum. In Appendix A.4, the matrix image of $B/2$ is the conventional normalized density matrix. The two outcome probabilities are already contained in B :

$$p_{\pm}(\mathbf{u}) = \text{sc}(\Pi_{\pm}B) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (8.27)$$

For a spatial Pauli wavefunction, the directly computable global probabilities are instead the density-weighted integrals

$$p_{\pm}(\mathbf{u}) = \frac{\int \text{sc}(\psi^{\text{H}}\Pi_{\pm}\psi) \, \text{d}^3\mathbf{r}}{\int \text{sc}(\psi^{\text{H}}\psi) \, \text{d}^3\mathbf{r}}. \quad (8.28)$$

More generally, let $Z' = a + \mathbf{b} \cdot \boldsymbol{\sigma}$ be Hermitian, with $a \in \mathbb{R}$, $\mathbf{b} \in \mathbb{R}^3$, and $\mathbf{b} \neq 0$, and take $\mathbf{u} = \mathbf{b}/\|\mathbf{b}\|$. Its spectral form and expectation are

$$Z' = (a + \|\mathbf{b}\|)\Pi_+ + (a - \|\mathbf{b}\|)\Pi_-, \quad (8.29)$$

$$\langle Z' \rangle_{\psi} = \text{sc}(Z'B) = a + \mathbf{b} \cdot \mathbf{s} \quad (8.30)$$

$$= (a + \|\mathbf{b}\|)p_+(\mathbf{u}) + (a - \|\mathbf{b}\|)p_-(\mathbf{u}). \quad (8.31)$$

Thus the same two projectors give the complete measurement rule for every Hermitian paravector observable. Cyclicity also gives the adjoint relation

$$\langle \psi' | Z'\psi \rangle = \langle Z'^{\text{H}}\psi' | \psi \rangle, \quad (8.32)$$

so left multiplication by Z' has adjoint left multiplication by Z'^{H} . Appendix A.4 shows that the fixed right projection gives one conventional two-component Pauli column, while the left projections give its ordinary spin-measurement components. Applying the same fixed right projection to both entries of the Dirac block gives the conventional four complex components.

8.3 Spin carried by paravector components

Here R is the rotation rotor defined in Section 9, a is a complex scalar amplitude, $\psi\Pi_3 = \psi$ is a minimal-ideal amplitude, and $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ is an unrestricted complex paravector amplitude. Spin is determined by how an amplitude transforms under rotations, not merely by how many coefficients are used to write it. The scalar, minimal-ideal, and vector carriers transform differently.

A complex scalar amplitude has no spatial direction and is unchanged by the rotor:

$$a' = a. \quad (8.33)$$

It therefore carries spin zero. The Klein–Gordon field below is of this type; its single complex coefficient is proportional to the algebra identity.

A minimal-ideal amplitude contains two complex coefficients. Its associated Pauli column transforms by one-sided rotor multiplication, which is represented intrinsically by

$$\psi' = R^H \psi. \quad (8.34)$$

This choice gives the same active rotation as the vector sandwich $R^H Z R$ used below. It also preserves the fixed ideal because $(R^H \psi)\Pi_3 = R^H \psi$. Since R contains the half-angle, a full 2π rotation gives

$$\psi' = -\psi. \quad (8.35)$$

This is the spin-one-half carrier. A Dirac amplitude consists of two such carriers coupled by the mass term; the pair still describes spin one-half, not spin zero plus spin one.

A pure vector amplitude $\mathbf{V} \cdot \boldsymbol{\sigma}$ transforms by the two-sided sandwich

$$\mathbf{V}' \cdot \boldsymbol{\sigma} = R^H (\mathbf{V} \cdot \boldsymbol{\sigma}) R. \quad (8.36)$$

The two rotor signs cancel under a 2π rotation, so this three-component carrier returns to itself and carries spin one. The electromagnetic field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is a complex pure vector of this type. The full real Maxwell field has the two physical helicities. After a frequency split, the self-dual field F carries one helicity on a positive-frequency branch, while its Hermitian-conjugate sector supplies the other. The three algebraic vector coefficients should not be confused with three independent photon polarizations. The classical Maxwell field carries energy density and Poynting flux as derived in Section 7; it is not by itself a normalized single-photon position wavefunction. More generally, a pure vector has the rotation content of spin one but is not automatically a physical particle field; a massive spin-one theory additionally requires its field equation and constraint and has three physical polarizations.

Consequently, under the two-sided sandwich an unrestricted paravector $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ is reducible: its scalar component carries spin zero and its vector component carries spin one. The transformation law is essential to this statement. Under one-sided multiplication, the matrix image of an unrestricted paravector consists of two Pauli columns and carries two copies of the spin-one-half representation. The fixed right projection selects one of those copies; it does not select scalar and vector coefficients.

Under Lorentz boosts the distinction becomes sharper. A scalar remains a scalar, the electromagnetic pure vector obeys $F' = L^{-1} F L$, and the two minimal-ideal entries of a Dirac state use the opposite factors L and L^{-1} . When $m = 0$ the two Dirac equations decouple into the two chiral spin-one-half equations. For a positive-energy massless mode with $\mathbf{u} = \mathbf{p}/\|\mathbf{p}\|$, the free equation gives

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\Psi_1 = -\Psi_1, \quad (\mathbf{u} \cdot \boldsymbol{\sigma})\Psi_2 = \Psi_2. \quad (8.37)$$

Thus the two entries become opposite helicities in the massless limit. They are not, by themselves, the spin-up/down or particle/antiparticle entries. The mass term couples them when $m \neq 0$.

8.4 Energy and momentum as differential operators

Recall the real space–time paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ and the real four-momentum $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$. Their invariant scalar pairing is

$$f(X) := \text{sc}(\text{adj}(P) X) = Et - \mathbf{p} \cdot \mathbf{r}. \quad (8.38)$$

The corresponding scalar plane wave is

$$w(X) := \exp(-\text{i}f(X)) = \exp(-\text{i}Et + \text{i}\mathbf{p} \cdot \mathbf{r}). \quad (8.39)$$

Its response to time and space translations gives

$$\text{i}\partial_t w = Ew, \quad (8.40)$$

$$-\text{i}\partial_{\mathbf{r}} w = \mathbf{p}w. \quad (8.41)$$

Thus energy and momentum are not separate substitutions added to the algebra: they are the scalar and vector coefficients of one paravector differential eigenvalue. Indeed,

$$\text{sc}(\text{i}\partial^*) = \text{i}\partial_t, \quad \text{vec}(\text{i}\partial^*) = -\text{i}\partial_{\mathbf{r}}, \quad (8.42)$$

$$\text{i}\partial^* w = Pw, \quad \text{i}\partial w = \text{adj}(P) w. \quad (8.43)$$

Here coefficient extraction from a differential paravector is understood before it acts. On the spatially integrated pairing, the three components of $-\text{i}\partial_{\mathbf{r}}$ are Hermitian when boundary terms vanish. The reality of an energy spectrum follows after the wave equation has been put in the Hermitian evolution form $\text{i}\partial_t \Psi = H\Psi$.

The particle action gives canonical momentum $\mathbf{p} + q\mathbf{A}$ and canonical energy $E + qV$. Identifying these translation generators with $-\text{i}\partial_{\mathbf{r}}$ and $\text{i}\partial_t$ respectively gives the kinetic operators below and directly links minimal coupling to the classical action. In an electromagnetic potential the gauge-covariant, or kinetic, operators are

$$E := \text{i}\partial_t - qV, \quad \mathbf{p} := -\text{i}\partial_{\mathbf{r}} - q\mathbf{A}, \quad (8.44)$$

$$E + \mathbf{p} \cdot \boldsymbol{\sigma} = \text{i}\partial^* - q\Phi, \quad E - \mathbf{p} \cdot \boldsymbol{\sigma} = \text{i}\partial - q\Phi^*. \quad (8.45)$$

The last two expressions have the same coefficient pattern as P and $\text{adj}(P)$. They are nevertheless differential operators. In a field their components do not commute, so their ordered products must be retained; one may not replace such a product by the ordinary paravector determinant. For the free operators all components commute and

$$(\text{i}\partial)(\text{i}\partial^*) = (\text{i}\partial^*)(\text{i}\partial) = -\square. \quad (8.46)$$

The electromagnetic remainder produced when the covariant orderings are interchanged is derived explicitly below in the interacting factorization.

8.5 Klein–Gordon equation

Let ψ be a complex scalar amplitude. For a free momentum $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$, the mass shell is the single algebraic statement

$$\det(P) = m^2. \quad (8.47)$$

Equations (8.43) and (8.46) turn this determinant polynomial into

$$((i\partial)(i\partial^*) - m^2) \psi = -(\square + m^2) \psi = 0. \quad (8.48)$$

Thus the Klein–Gordon equation is

$$(\square + m^2) \psi = 0. \quad (8.49)$$

In an electromagnetic potential use the covariant operators in equation (8.45). The spin-zero equation is the scalar part of either ordered paravector product:

$$(-E^2 + \mathbf{p}^2 + m^2) \psi = (-\text{sc}((E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + \mathbf{p} \cdot \boldsymbol{\sigma})) + m^2) \psi = 0. \quad (8.50)$$

The vector part is not discarded accidentally: it is absent from the scalar equation but becomes the electromagnetic spin coupling when the Dirac operator is squared.

For $m > 0$, the conserved Klein–Gordon current has the intrinsic paravector form

$$J := \frac{1}{m} \text{re}(\psi^H (E + \mathbf{p} \cdot \boldsymbol{\sigma}) \psi) = j + \mathbf{j} \cdot \boldsymbol{\sigma}. \quad (8.51)$$

The covariant equation and its Hermitian form give

$$\text{sc}(\partial J) = \partial_t j + \partial_{\mathbf{r}} \cdot \mathbf{j} = 0. \quad (8.52)$$

For the free branches $\exp(-if)$ and $\exp(+if)$, respectively,

$$J = +\frac{\psi^H \psi}{m} P, \quad J = -\frac{\psi^H \psi}{m} P. \quad (8.53)$$

Its time component is therefore positive on the positive-frequency branch and negative on the negative-frequency branch. Consequently j is a signed conserved density and qj is the electric charge density; j is not a positive probability density on an arbitrary Klein–Gordon solution. A one-particle probability interpretation requires restriction to a positive-frequency sector, where the integrated Klein–Gordon pairing is positive; its local density need not be pointwise positive for a superposition. The complete particle–antiparticle interpretation is the quantized scalar field. This is why the positive pairing used for Pauli and Dirac wavefunctions was not assigned to the relativistic scalar field.

8.6 Spin-zero low-energy limit

Let $E = i\partial_t - qV$ and $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$, with $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, and denote the relativistic scalar solution by ψ' . Separate its rest-energy phase by $\psi' = \exp(-imt)\psi$. The exact equation for the slowly varying amplitude becomes

$$(-E^2 - 2mE + \mathbf{p}^2) \psi = 0 \quad (8.54)$$

$$E\psi = \frac{\mathbf{p}^2 - E^2}{2m} \psi. \quad (8.55)$$

For a free positive-frequency particle the operators commute and the exact residual-energy branch is

$$E = \sqrt{\mathbf{p}^2 + m^2} - m. \quad (8.56)$$

Its expansion gives

$$E = \frac{\mathbf{p}^2}{2m} - \frac{\mathbf{p}^4}{8m^3} + \cdots. \quad (8.57)$$

More generally, neglecting E^2 at leading nonrelativistic order in the covariant equation gives the Schrödinger equation

$$i\partial_t\psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi. \quad (8.58)$$

8.7 Dirac equation in block form

Let Z be a paravector, $\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}$, and $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ when a field is present. Here $\boldsymbol{\Psi} = \{\Psi_1, \Psi_2\}^\top$ is a two-entry block whose entries belong to the same fixed minimal left ideal:

$$\Psi_1\Pi_3 = \Psi_1, \quad \Psi_2\Pi_3 = \Psi_2. \quad (8.59)$$

The existing Hermitian operation extends to blocks by transposing the block and applying H to each entry. Thus, if a block has entries Z_{nm} , then its Hermitian block has entries Z_{mn}^H . This is only the block extension of the operation already defined in Section 2. Sigma conjugation of a block acts entrywise without transposing the block. The componentwise Dirac pairing is

$$\langle \boldsymbol{\Psi}' | \boldsymbol{\Psi} \rangle := \sum_{k=1}^2 \text{sc}(\Psi_k'^H \Psi_k), \quad (8.60)$$

with a spatial integral for wavefunctions. The corresponding local Dirac norm density is

$$\|\boldsymbol{\Psi}\|^2 = \text{sc}(\Psi_1^H \Psi_1) + \text{sc}(\Psi_2^H \Psi_2). \quad (8.61)$$

Define

$$\mathbf{W}(Z) := \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix}. \quad (8.62)$$

It satisfies

$$\mathbf{W}(Z)^2 = \begin{Bmatrix} Z \text{adj}(Z) & 0 \\ 0 & \text{adj}(Z) Z \end{Bmatrix} = \det(Z) \mathbf{I} \quad (8.63)$$

$$\mathbf{W}(Z)^* = \mathbf{W}(Z^*). \quad (8.64)$$

For the free field, define the Dirac operator and its mass conjugate:

$$\mathbf{D} := \mathbf{W}(i\partial) - m\mathbf{I} \quad (8.65)$$

$$\mathbf{D}' := \mathbf{W}(i\partial) + m\mathbf{I}. \quad (8.66)$$

The Dirac equation is

$$\mathbf{D}\boldsymbol{\Psi} = 0. \quad (8.67)$$

Since $\mathbf{W}(i\partial)^2 = -\square\mathbf{I}$, multiplication by the mass conjugate gives

$$\mathbf{D}'\mathbf{D} = \mathbf{W}(i\partial)^2 - m^2\mathbf{I} = -(\square + m^2)\mathbf{I}. \quad (8.68)$$

Consequently, $\mathbf{D}\Psi = 0$ implies $(\square + m^2)\Psi = 0$, so every component of the Dirac block satisfies the Klein–Gordon equation. This factorization is the direct connection between the first-order Dirac equation and the relativistic mass-shell equation. For $\Psi = \{\Psi_1, \Psi_2\}^\top$ it reads

$$i\partial\Psi_2 - m\Psi_1 = 0 \quad (8.69)$$

$$i\partial^*\Psi_1 - m\Psi_2 = 0. \quad (8.70)$$

Conversely, for $m \neq 0$, any scalar solution ψ' of $(\square + m^2)\psi' = 0$ gives a Dirac solution

$$\Psi = \left\{ \psi'\Pi_3, \frac{i}{m}(\partial^*\psi')\Pi_3 \right\}^\top. \quad (8.71)$$

Indeed, the second component equation holds by definition, and the first reduces to $-(\square + m^2)\psi'/m = 0$. More generally, for constant complex scalars a and b , the same scalar solution gives

$$\Psi = \frac{1}{m} \left\{ \begin{matrix} ma + ib\partial \\ ia\partial^* + mb \end{matrix} \right\} \psi'\Pi_3. \quad (8.72)$$

Substitution into the two component equations leaves respectively $-a(\square + m^2)\psi'/m$ and $-b(\square + m^2)\psi'/m$, so both vanish. This is a two-constant family generated from one Klein–Gordon solution, not a claim that one scalar solution parametrizes every Dirac solution. With a potential, the gauge-covariant form is

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (8.73)$$

It is invariant under $\Phi' = \Phi + \partial^*\lambda$ and $\Psi = \exp(iq\lambda)\Psi'$.

Isolating the time derivative in the two paravector equations gives the ordinary evolution form

$$i\partial_t\Psi = H\Psi, \quad H := qV\mathbf{I} + \mathbf{W}(1)(m\mathbf{I} + \mathbf{W}(\mathbf{p} \cdot \boldsymbol{\sigma})) = qV\mathbf{I} + \left\{ \begin{matrix} -\mathbf{p} \cdot \boldsymbol{\sigma} & m \\ m & \mathbf{p} \cdot \boldsymbol{\sigma} \end{matrix} \right\}, \quad (8.74)$$

where $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$. For real V and $\mathbf{A}$, suitable boundary conditions make $\mathbf{p}$ and hence H Hermitian with respect to the integrated Dirac pairing. Therefore equation (8.74) conserves the spatial integral of the positive density in equation (8.61). This operator statement includes the integration by parts implicit in spatial derivatives; it is stronger than applying the algebraic H pointwise.

In the rest of this subsection, J denotes the Dirac probability-current paravector rather than the electromagnetic source. It has the intrinsic form

$$J := \text{adj}(\Psi_1\Psi_1^H) + \Psi_2\Psi_2^H = j + \mathbf{j} \cdot \boldsymbol{\sigma}. \quad (8.75)$$

Its scalar and vector components are

$$j = \text{sc}(\Psi_1^H\Psi_1) + \text{sc}(\Psi_2^H\Psi_2), \quad (8.76)$$

$$\mathbf{j} = -\text{vec}(\Psi_1\Psi_1^H) + \text{vec}(\Psi_2\Psi_2^H). \quad (8.77)$$

Here j is the positive equal-time density. The Dirac equation and its Hermitian form give

$$\text{sc}(\partial J) = \partial_t j + \partial_{\mathbf{r}} \cdot \mathbf{j} = 0. \quad (8.78)$$

Thus the spatial integral of j is conserved. The ideal condition makes each term in J a future-null Hermitian paravector. Their sum is therefore future causal, and direct use of the determinant gives

$$J^H = J, \quad \det(J) = j^2 - \mathbf{j}^2 = 4 \left\| \text{sc}(\Psi_1^H \Psi_2) \right\|^2 \geq 0. \quad (8.79)$$

Consequently $\|\mathbf{j}\| \leq j$; where $j \neq 0$, the ratio $\mathbf{j}/j$ is the probability-flow velocity. It is not the spin direction.

The full paravector J , not the scalar component by itself, has a Lorentz-covariant transformation law. For a wavefunction normalized on a constant-time slice,

$$\int j \, d^3\mathbf{r} = 1. \quad (8.80)$$

More generally, if N is the future unit-normal paravector to a spacelike hypersurface and dA its invariant volume element, the covariant normalization is

$$\int \text{sc}(\text{adj}(N) J) \, dA = 1. \quad (8.81)$$

For a constant-time surface, $N = 1$ and this reduces to the preceding spatial integral. The corresponding conventional column formulas are given in [Appendix A.4](#).

8.8 Schrödinger–Pauli limit

Recall $\Psi = \{\Psi_1, \Psi_2\}^T$, $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$, and $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Each relativistic Dirac component contains the rapid phase associated with the rest energy. For $n = 1, 2$, separate it from the slowly varying component by

$$\Psi'_n = \exp(-imt)\Psi_n. \quad (8.82)$$

For the covariant operators $E = i\partial_t - qV$ and $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$, the exponential is spatially constant and therefore

$$E\Psi'_n = \exp(-imt)(E + m)\Psi_n, \quad (8.83)$$

$$\mathbf{p}\Psi'_n = \exp(-imt)\mathbf{p}\Psi_n. \quad (8.84)$$

After cancelling the common exponential, the two Dirac equations become

$$(E + m - \mathbf{p} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 = 0, \quad (8.85)$$

$$(E + m + \mathbf{p} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 = 0. \quad (8.86)$$

Eliminating Ψ_1 gives the exact second-order equation

$$(E^2 + 2mE - \mathbf{p}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma})\Psi_2 = 0. \quad (8.87)$$

Under the usual low-energy, slowly varying-field power counting, the E^2 term and the electric-field term contribute beyond the leading nonrelativistic approximation. The expansion is in energy and momentum relative to m ; it is not an expansion in small m . Retaining the leading terms gives

$$E\Psi_2 = \frac{\mathbf{p}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\Psi_2. \quad (8.88)$$

At this order Ψ_1 and Ψ_2 have the same leading rest-frame amplitude. The conventionally normalized large component is their sum divided by $\sqrt{2}$; either common component obeys the same leading equation after a constant normalization. Writing the faithful two-component column as ψ and substituting the covariant operators gives the Pauli equation

$$i\partial_t\psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right) \psi. \quad (8.89)$$

In this column formula, each sigma-algebra element acts through its faithful matrix image. Since the physical spin is one half of $\boldsymbol{\sigma}$, the magnetic term is $-(q/m)\mathbf{B} \cdot (\boldsymbol{\sigma}/2)$. Thus minimal Dirac coupling gives magnetic moment q/m times the spin, or $g = 2$, before radiative corrections. Using the spin projectors, $\psi = \psi_+ + \psi_-$ with $\psi_{\pm} = \Pi_{\pm}\psi$. For a uniform static magnetic field, $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$, the sectors decouple:

$$i\partial_t\psi_{\pm} = \left(\frac{\mathbf{p}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \psi_{\pm}. \quad (8.90)$$

For a varying field direction the sectors are coupled. With H denoting the Pauli Hamiltonian, direct differentiation gives

$$i\partial_t\psi_{\pm} = i(\partial_t\Pi_{\pm})\psi + \Pi_{\pm}H\psi \quad (8.91)$$

$$= H\psi_{\pm} + (i\partial_t\Pi_{\pm} + [\Pi_{\pm}, H])\psi. \quad (8.92)$$

Because the field term commutes with its own instantaneous projectors, this is

$$i\partial_t\psi_{\pm} = H\psi_{\pm} + \mathbf{S}_{\pm} \quad (8.93)$$

$$\mathbf{S}_{\pm} = \left(i\partial_t\Pi_{\pm} + \frac{[\Pi_{\pm}, \mathbf{p}^2]}{2m} \right) \psi. \quad (8.94)$$

8.9 Interacting operators and factorization

Recall $E = i\partial_t - qV$, $\mathbf{p} = -i\partial_{\mathbf{r}} - q\mathbf{A}$, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, with $\boldsymbol{\Psi} = \{\Psi_1, \Psi_2\}^T$. The covariant energy and momentum operators do not commute. Direct application to a test wavefunction gives

$$[E, p_m] = iqE_m \quad (8.95)$$

$$[p_m, p_n] = iq(\partial_m A_n - \partial_n A_m) \quad (8.96)$$

$$\mathbf{p} \times \mathbf{p} = iq\mathbf{B}. \quad (8.97)$$

Therefore

$$(E + \boldsymbol{\sigma} \cdot \mathbf{p})(E - \boldsymbol{\sigma} \cdot \mathbf{p}) = E^2 - \mathbf{p}^2 - iqF, \quad (8.98)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{p})(E + \boldsymbol{\sigma} \cdot \mathbf{p}) = E^2 - \mathbf{p}^2 - iqF^*, \quad (8.99)$$

$$E + \boldsymbol{\sigma} \cdot \mathbf{p} = i\partial^* - q\Phi, \quad (8.100)$$

$$E - \boldsymbol{\sigma} \cdot \mathbf{p} = i\partial - q\Phi^*. \quad (8.101)$$

Equivalently, scalar and vector extraction expose the two distinct pieces:

$$\text{sc}((E + \mathbf{p} \cdot \boldsymbol{\sigma})(E - \mathbf{p} \cdot \boldsymbol{\sigma})) = E^2 - \mathbf{p}^2, \quad (8.102)$$

$$\text{sc}((E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + \mathbf{p} \cdot \boldsymbol{\sigma})) = E^2 - \mathbf{p}^2, \quad (8.103)$$

$$\text{vec}((E + \mathbf{p} \cdot \boldsymbol{\sigma})(E - \mathbf{p} \cdot \boldsymbol{\sigma})) \cdot \boldsymbol{\sigma} = -iqF, \quad (8.104)$$

$$\text{vec}((E - \mathbf{p} \cdot \boldsymbol{\sigma})(E + \mathbf{p} \cdot \boldsymbol{\sigma})) \cdot \boldsymbol{\sigma} = -iqF^*. \quad (8.105)$$

The Klein–Gordon equation retains the common scalar part. Dirac factorization retains the full ordered paravectors, so their vector remainders are precisely the spin–field couplings. This is the algebraic reason spin appears when the first-order equation is squared. Expanding the minimally coupled Klein–Gordon operator gives the fully explicit result

$$(\square + iqS - q^2 \det(\Phi) + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})) \psi = 0. \quad (8.106)$$

The two orderings satisfy

$$0 = ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2) \Psi_1 = (-E^2 + \mathbf{p}^2 + m^2 + iqF^*)\Psi_1, \quad (8.107)$$

$$0 = ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2) \Psi_2 = (-E^2 + \mathbf{p}^2 + m^2 + iqF)\Psi_2. \quad (8.108)$$

These follow from the same mass-conjugate factorization used for the free Dirac equation. With the potential included, the two block operators become

$$\mathbf{D} := \mathbf{W}(\mathrm{i}\partial - q\Phi^*) - m\mathbf{I}, \quad (8.109)$$

$$\mathbf{D}' := \mathbf{W}(\mathrm{i}\partial - q\Phi^*) + m\mathbf{I}. \quad (8.110)$$

Then

$$\mathbf{D}'\mathbf{D} = - \begin{Bmatrix} -E^2 + \mathbf{p}^2 + m^2 + iqF^* & 0 \\ 0 & -E^2 + \mathbf{p}^2 + m^2 + iqF \end{Bmatrix}. \quad (8.111)$$

The electromagnetic field therefore prevents the interacting product from being proportional to the block identity: its two components acquire the distinct spin couplings iqF^* and iqF . In the free limit $q = 0$, both couplings vanish and $-E^2 + \mathbf{p}^2 = \square$, so

$$\mathbf{D}'\mathbf{D} = -(\square + m^2)\mathbf{I}. \quad (8.112)$$

Thus the free Dirac equation $\mathbf{D}\Psi = 0$ recovers the Klein–Gordon equation on both components of the Dirac block.

The same formulas close the classical field–matter system without leaving the paravector algebra. For one charged Dirac field, take

$$F^* = \partial\Phi - \mathrm{sc}(\partial\Phi), \quad D = qJ, \quad \partial F = D^*, \quad (8.113)$$

together with the interacting Dirac equation (8.73). Then $\mathrm{sc}(\partial D) = q\mathrm{sc}(\partial J) = 0$ is exactly the source consistency required by Maxwell’s equation. Replacing the Dirac current by the Klein–Gordon current (8.51) gives the corresponding classical scalar system. These are the Maxwell–Dirac and Maxwell–Klein–Gordon equations; quantized photons, particle creation, and radiative corrections require quantum field theory.

8.10 Spin, energy sign, and probabilities

The Pauli-state paravector B and the spin probabilities $p_{\pm}(\mathbf{u}) = \mathrm{sc}(\Pi_{\pm}B)$ were constructed intrinsically in equations (8.25) and (8.27). For a Dirac block, the corresponding local spin-state paravector is

$$B := \frac{\Psi_1\Psi_1^{\mathrm{H}} + \Psi_2\Psi_2^{\mathrm{H}}}{j} = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}, \quad j \neq 0. \quad (8.114)$$

It satisfies $B^{\mathrm{H}} = B$, $\mathrm{sc}(B) = 1$, and $\det(B) = 1 - \mathbf{s}^2 \geq 0$. A Dirac block can therefore have $\mathbf{s}^2 < 1$ even when the full block is a pure state, because its two chiral entries need not have the same spin

direction. It must not be confused with the current J : the adjugate on the first term of J makes its vector part probability flow, whereas the same-sign sum in B gives spin. The spin projectors act identically from the left on both entries,

$$\Pi_{\pm} \Psi := \{\Pi_{\pm} \Psi_1, \Pi_{\pm} \Psi_2\}^T, \quad (8.115)$$

and the local outcome fractions are

$$\text{sc}(\Pi_{\pm} B) = \frac{\text{sc}(\Psi_1^H \Pi_{\pm} \Psi_1) + \text{sc}(\Psi_2^H \Pi_{\pm} \Psi_2)}{j} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (8.116)$$

For a spatial wavefunction, the directly computable probability is

$$p_{\pm}(\mathbf{u}) = \frac{\int \left(\text{sc}(\Psi_1^H \Pi_{\pm} \Psi_1) + \text{sc}(\Psi_2^H \Pi_{\pm} \Psi_2) \right) d^3 \mathbf{r}}{\int j d^3 \mathbf{r}}. \quad (8.117)$$

For a normalized state the denominator is one. If a pure state is aligned with $\mathbf{u}$ and $\mathbf{u} \cdot \mathbf{u}' = \cos \alpha$, a measurement along $\mathbf{u}'$ gives $\cos^2(\alpha/2)$ and $\sin^2(\alpha/2)$.

For the free field, the Hamiltonian follows directly from the paravector block function:

$$H = \mathbf{W}(1) (m \mathbf{I} + \mathbf{W}(\mathbf{p} \cdot \boldsymbol{\sigma})) = \begin{Bmatrix} -\mathbf{p} \cdot \boldsymbol{\sigma} & m \\ m & \mathbf{p} \cdot \boldsymbol{\sigma} \end{Bmatrix}. \quad (8.118)$$

At fixed real momentum, let $E = \sqrt{\mathbf{p}^2 + m^2} > 0$. Then

$$H^H = H, \quad H^2 = E^2 \mathbf{I}. \quad (8.119)$$

The Hermitian energy projectors are therefore

$$\mathbf{Q}_{\pm} := \frac{1}{2} \left(\mathbf{I} \pm \frac{H}{E} \right). \quad (8.120)$$

They satisfy $\mathbf{Q}_{\pm}^H = \mathbf{Q}_{\pm}$, $\mathbf{Q}_{\pm}^2 = \mathbf{Q}_{\pm}$, $\mathbf{Q}_+ \mathbf{Q}_- = 0$, and $\mathbf{Q}_+ + \mathbf{Q}_- = \mathbf{I}$. Their directly computable probabilities are

$$\frac{\|\mathbf{Q}_+ \Psi\|^2}{\|\Psi\|^2}, \quad \frac{\|\mathbf{Q}_- \Psi\|^2}{\|\Psi\|^2}. \quad (8.121)$$

For a wave packet, let $\Psi(\mathbf{p})$ denote its momentum-space block and use $H(\mathbf{p})$ and $E(\mathbf{p}) = \sqrt{\mathbf{p}^2 + m^2}$ in the preceding formulas. For example, the positive-energy probability is

$$\frac{\int \left\| \frac{1}{2} (\mathbf{I} + H(\mathbf{p})/E(\mathbf{p})) \Psi(\mathbf{p}) \right\|^2 d^3 \mathbf{p}}{\int \|\Psi(\mathbf{p})\|^2 d^3 \mathbf{p}}, \quad (8.122)$$

with the sign reversed for the negative-energy probability. In position space $E = \sqrt{-\partial_{\mathbf{r}}^2 + m^2}$, so these free energy projectors are nonlocal.

For $m > 0$, the covariant mass-shell idempotents have a different role. Put

$$U = \frac{P}{m} = \frac{E + \mathbf{p} \cdot \boldsymbol{\sigma}}{m}. \quad (8.123)$$

The mass shell gives $\det(U) = \det(P)/m^2 = 1$, and therefore $\mathbf{W}(\text{adj}(U))^2 = \mathbf{I}$. Hence the following expressions are idempotents. For the phase $w = \exp(-if)$, the free Dirac equation requires the constant amplitude to lie in the +1 eigenspace of $\mathbf{W}(\text{adj}(U))$. For the conjugate phase $w^H = \exp(+if)$, it lies in the -1 eigenspace. Hence

$$\frac{1}{2}(\mathbf{I} + \mathbf{W}(\text{adj}(U))), \quad \frac{1}{2}(\mathbf{I} - \mathbf{W}(\text{adj}(U))) \quad (8.124)$$

are the covariant positive- and negative-frequency amplitude idempotents. The conjugate phase has canonical spatial momentum $-\mathbf{p}$. To compare its amplitude with the Hamiltonian projector $\mathbf{Q}_-(\mathbf{p})$ at a fixed momentum label $\mathbf{p}$, first replace $\mathbf{p}$ by $-\mathbf{p}$ in the future momentum and then take the conjugate phase. Equivalently, $\mathbf{Q}_-(\mathbf{p})$ acts on the negative-frequency phase $\exp(+iEt + i\mathbf{p} \cdot \mathbf{r})$, whereas the displayed w^H corresponds directly to $\mathbf{Q}_-(\mathbf{p})$. They are not Hermitian under the positive componentwise pairing away from rest, because

$$\mathbf{W}(\text{adj}(U))^H = \mathbf{W}(U) \neq \mathbf{W}(\text{adj}(U)) \quad (\mathbf{p} \neq 0). \quad (8.125)$$

Their squared component norms are therefore not probabilities; the Hermitian projectors $\mathbf{Q}_\pm$ are used for that purpose.

For a massive free field, the four joint spin-energy basis states are most transparent at rest. When $\mathbf{p} = 0$,

$$H = m \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}. \quad (8.126)$$

Positive energy means $\Psi_2 = \Psi_1$, while negative energy means $\Psi_2 = -\Psi_1$. With the spin basis from equation (8.13), the four normalized rest states are

$$\frac{1}{\sqrt{2}}\{e_+, e_+\}^\top, \quad \frac{1}{\sqrt{2}}\{e_-, e_-\}^\top, \quad (8.127)$$

$$\frac{1}{\sqrt{2}}\{e_+, -e_+\}^\top, \quad \frac{1}{\sqrt{2}}\{e_-, -e_-\}^\top. \quad (8.128)$$

Thus the choice e_+ or e_- is spin up or down, while the relative sign between the two chiral entries is the energy sign.

Negative energy is not by itself an antimatter label. The existing sigma conjugation gives a charge-reversed solution without introducing another conjugation operation. If Ψ has charge q , define

$$\Psi_c := \{\Psi_2^* \sigma_1, -\Psi_1^* \sigma_1\}^\top. \quad (8.129)$$

Since $\Pi_3^* \sigma_1 = \sigma_1 \Pi_3$, both entries of Ψ_c remain in the fixed minimal left ideal. Applying $*$ to the two coupled Dirac equations shows that Ψ_c satisfies the same equations with q replaced by $-q$. Sigma conjugation also reverses the scalar plane-wave phase, so it maps a negative-frequency charge- q mode to a positive-frequency charge- $-q$ mode. Since $\Pi_\pm^* = \Pi_\mp$, the raw Pauli-projector label also reverses under this map; the physical antiparticle spin is determined by applying the desired spin projector to the charge-reversed state. After field quantization the coefficients of the original negative-frequency modes become creation amplitudes for positive-energy antiparticles. Thus $\mathbf{Q}_-$ is a negative-frequency projector for the one-particle charge- q equation, not an antimatter-probability projector. For a free field, proper orthochronous Lorentz transformations preserve the future and past mass shells and do not mix these two frequency branches.

Finally, $\Pi_\pm$ describes the canonical Pauli spin in the chosen inertial frame. Spin and energy give simultaneous alternatives only when their projectors commute. For a fixed free momentum this

occurs for any spin axis at rest, and away from rest for $\mathbf{u}$ collinear with $\mathbf{p}$. Taking $\mathbf{u} = \mathbf{p}/\|\mathbf{p}\|$ gives the usual helicity labels; reversing the axis reverses those labels. A covariant arbitrary rest-frame spin axis would require a momentum-dependent boosted projector, so it is not represented by simply holding $\mathbf{u}$ fixed in $\Pi_{\pm}$. For $a, b \in \{+1, -1\}$, write

$$\mathbf{Q}_a := \frac{1}{2}(\mathbf{I} + aH/E), \quad \Pi_b := \frac{1}{2}(1 + b\mathbf{u} \cdot \boldsymbol{\sigma}). \quad (8.130)$$

Here Π_b acts componentwise on both Dirac entries, so $[\mathbf{Q}_a, \Pi_b]$ is a commutator of block operators. When $[\mathbf{Q}_a, \Pi_b] = 0$, the four joint probabilities are

$$p_{ab} = \frac{\|\mathbf{Q}_a \Pi_b \Psi\|^2}{\|\Psi\|^2}, \quad \sum_{a,b} p_{ab} = 1. \quad (8.131)$$

They describe positive or negative energy combined with spin up or down. For a wave packet, a fixed axis gives joint projectors only when this commutation condition holds throughout its momentum support.

If the projectors do not commute, the question must specify the measurement order. Energy followed by spin and spin followed by energy give respectively

$$\frac{\|\Pi_b \mathbf{Q}_a \Psi\|^2}{\|\Psi\|^2}, \quad \frac{\|\mathbf{Q}_a \Pi_b \Psi\|^2}{\|\Psi\|^2}, \quad (8.132)$$

and need not agree. The conditional spin probability after an energy result a is $\|\Pi_b \mathbf{Q}_a \Psi\|^2 / \|\mathbf{Q}_a \Psi\|^2$, provided $\|\mathbf{Q}_a \Psi\|^2 \neq 0$. For a static external field, energy sectors are instead defined by the spectrum of the full Hermitian Hamiltonian (8.74). A time-dependent field need not preserve any fixed positive-negative split; its mixing of the two sectors requires field quantization for a complete pair-production interpretation.

9 Rotations and Lorentz boosts

Throughout this section, the transformation factor T is defined from a positive generator. Thus $T = R$ for rotations and $T = L$ for boosts. The opposite action is written with T^{-1} ; it is not introduced by redefining T with a negative exponential.

9.1 Rotors

Let $\theta \in \mathbb{R}$ be the rotation angle, $\mathbf{u} \in \mathbb{R}^3$ a unit axis, $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ a spacetime paravector, Z a paravector, and $\psi \Pi_3 = \psi$ a minimal-ideal amplitude. The rotation is generated by

$$R := \exp\left(i\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} + \mathbf{u} \cdot i\boldsymbol{\sigma} \sin \frac{\theta}{2} \quad (9.1)$$

$$R^{-1} = R^H = \cos \frac{\theta}{2} - \mathbf{u} \cdot i\boldsymbol{\sigma} \sin \frac{\theta}{2}. \quad (9.2)$$

The sandwich action $X' = R^H X R$ fixes the scalar component and rotates the spatial vector according to Rodrigues' formula,

$$\mathbf{r}' = \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \theta)\mathbf{u}. \quad (9.3)$$

The half-angle explains the spinorial sign change $R(\theta + 2\pi) = -R(\theta)$. The corresponding column-spinor and density-matrix transformations are collected in Appendix A.4. For a spinor undergoing the same rotation, let $\psi_R := R^H \psi$. Then

$$\psi_R^H = \psi^H R \quad (9.4)$$

$$\langle Z \rangle_{\psi_R} = \frac{\text{sc}(\psi^H R Z R^H \psi)}{\text{sc}(\psi^H \psi)} \quad (9.5)$$

$$R^H = R^{-1}. \quad (9.6)$$

If the observable is rotated as $Z' = R^H Z R$, cyclicity gives

$$\langle Z' \rangle_{\psi_R} = \langle Z \rangle_{\psi}. \quad (9.7)$$

In particular, $R^H(\mathbf{r} \cdot \boldsymbol{\sigma})R = \mathbf{r}' \cdot \boldsymbol{\sigma}$ gives the same Rodrigues rotation displayed above. Both entries of a Dirac block transform by this same factor:

$$\Psi'_1 = R^H \Psi_1, \quad \Psi'_2 = R^H \Psi_2. \quad (9.8)$$

The current transforms by the same vector sandwich:

$$J' = R^H J R. \quad (9.9)$$

9.2 Boosts

Let $\mathbf{u} \in \mathbb{R}^3$ be a unit boost direction, $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ a spacetime paravector, $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ an electromagnetic field, and $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ the potential, $D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$ the electromagnetic source, and $\Psi = \{\Psi_1, \Psi_2\}^T$ a Dirac block. Let rapidity θ satisfy $\beta = \tanh \theta$ and $\gamma = \cosh \theta$. The boost is

$$L := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh \frac{\theta}{2} + \mathbf{u} \cdot \boldsymbol{\sigma} \sinh \frac{\theta}{2} \quad (9.10)$$

$$L^H = L \quad (9.11)$$

$$L^{-1} = L^* = \cosh \frac{\theta}{2} - \mathbf{u} \cdot \boldsymbol{\sigma} \sinh \frac{\theta}{2}. \quad (9.12)$$

Unlike a rotation, a boost acts oppositely on the two chiral entries of a Dirac block. In the following field laws the primed field is evaluated at $X' = L^{-1} X L^{-1}$ and the unprimed field at X . With the ordering used in the Dirac equation,

$$\Psi'_1(X') = L \Psi_1(X), \quad \Psi'_2(X') = L^{-1} \Psi_2(X). \quad (9.13)$$

Thus $L^{-1}\psi$ is the transformation of the second chiral carrier, while $L\psi$ is the transformation of the first. A single minimal-ideal amplitude has no unique boost law until its chirality is specified. The two chiral representations use the same rotation factor but opposite boost generators, as required by the mass coupling.

The similarity $L^{-1} Z L$ is a different operation: it preserves $\text{sc}(Z)$ and is not the spacetime-vector boost used here. The spacetime paravector therefore transforms by the Hermitian congruence $X' = L^{-1} X L^{-1}$, which yields the components $\mathbf{r}_{\parallel} = (\mathbf{r} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}$, $\mathbf{r}'_{\parallel} = (\mathbf{r}' \cdot \mathbf{u})\mathbf{u}$, and $\mathbf{r}'_{\perp} = \mathbf{r}' - \mathbf{r}'_{\parallel}$:

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (9.14)$$

$$\mathbf{r}'_{\parallel} = \gamma(\mathbf{r}_{\parallel} - \beta t \mathbf{u}), \quad (9.15)$$

$$\mathbf{r}'_{\perp} = \mathbf{r}_{\perp}. \quad (9.16)$$

It preserves $\det(X) = t^2 - \mathbf{r}^2$. Four-momentum has the same vector law,

$$P' = L^{-1} P L^{-1}, \quad \text{sc}(\text{adj}(P') X') = \text{sc}(\text{adj}(P) X), \quad (9.17)$$

so the plane-wave phase is Lorentz invariant. The electromagnetic field instead obeys $F'(X') = L^{-1} F(X) L$ for $F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma}$. Define $\mathbf{E}_{\parallel} = (\mathbf{E} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}_{\perp} = \mathbf{E} - \mathbf{E}_{\parallel}$ and likewise $\mathbf{E}'_{\parallel} = (\mathbf{E}' \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}'_{\perp} = \mathbf{E}' - \mathbf{E}'_{\parallel}$; define the four corresponding components of $\mathbf{B}$ and $\mathbf{B}'$ by the same formulas. Then

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad (9.18)$$

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \beta \mathbf{u} \times \mathbf{B}), \quad (9.19)$$

$$\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}, \quad (9.20)$$

$$\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \beta \mathbf{u} \times \mathbf{E}). \quad (9.21)$$

With the same convention,

$$\Phi'(X') = L^{-1} \Phi(X) L^{-1} \quad (9.22)$$

$$\partial' = L \partial L \quad (9.23)$$

$$\partial'^* = L^{-1} \partial^* L^{-1} \quad (9.24)$$

$$D' = L^{-1} D L^{-1} \quad (9.25)$$

$$D'^* = L D^* L. \quad (9.26)$$

Since L is constant and $LL^{-1} = 1$, the field equation is covariant:

$$\partial' F' = L(\partial F) L = L D^* L = D'^*. \quad (9.27)$$

The potential and derivative transformations likewise give

$$i\partial' - q\Phi'^* = L(i\partial - q\Phi^*)L, \quad (9.28)$$

$$i\partial'^* - q\Phi' = L^{-1}(i\partial^* - q\Phi)L^{-1}. \quad (9.29)$$

Together with equation (9.13), these identities make both coupled Dirac equations covariant. The current in equation (8.75) transforms with the spacetime-vector law,

$$J'(X') = L^{-1} J(X) L^{-1}. \quad (9.30)$$

If the hypersurface normal obeys the same vector law, $N' = L^{-1} N L^{-1}$, then

$$\text{sc}(\text{adj}(N') J') = \text{sc}(\text{adj}(N) J) \quad (9.31)$$

by adjugation and cyclicity. This is the local invariance behind the hypersurface normalization (8.81).

10 Flat spacetime in generalized coordinates

This section changes coordinates within flat Minkowski spacetime; it does not introduce spacetime curvature or a gravitational field. For a time-like curve, recall that $U = dX/ds$ represents proper velocity. Let the real scalar function f and real three-vector function $\mathbf{g}$ define a locally invertible

map from $(t, \mathbf{r})$ to the inertial components of X . If $x_0 = t$ and $x_n = r_n$ for $n = 1, 2, 3$, write $\partial_n := \partial/\partial x_n$. For $n, m \in \{0, 1, 2, 3\}$, the pulled-back metric coefficients are

$$g_{nm} := \partial_n f \partial_m f - \partial_n \mathbf{g} \cdot \partial_m \mathbf{g}. \quad (10.1)$$

The determinant of the Jacobian of $(f, \mathbf{g})$ must be nonzero for these functions to form a local coordinate chart. Although g_{nm} may vary with the coordinates, it is the pullback of the flat metric.

Let

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma} \quad (10.2)$$

$$dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}. \quad (10.3)$$

Writing $dX = A dt + \mathbf{B} \cdot d\mathbf{r}$ gives

$$A = \partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma} \quad (10.4)$$

$$\mathbf{B} = \partial_r f + \partial_r (\mathbf{g} \cdot \boldsymbol{\sigma}), \quad (10.5)$$

and the interval remains

$$ds^2 = dX dX^* = df^2 - d\mathbf{g}^2. \quad (10.6)$$

For a future-directed time-like curve with $df > 0$,

$$ds = \sqrt{1 - \mathbf{v}^2} df \quad (10.7)$$

$$\mathbf{v}^2 < 1. \quad (10.8)$$

Define the geometric velocity and Lorentz factor by

$$\mathbf{v} = \frac{d\mathbf{g}}{df} \quad (10.9)$$

$$\gamma = \frac{df}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}} \quad (10.10)$$

$$\frac{dX}{ds} = U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (10.11)$$

With geometric acceleration $\mathbf{a} = d\mathbf{v}/ds = \gamma d^2\mathbf{g}/df^2$,

$$\frac{dU}{ds} = \gamma^3(\mathbf{v} \cdot \mathbf{a}) + (\gamma\mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (10.12)$$

For inertial motion, $dU/ds = 0$. Its scalar component gives $\mathbf{v} \cdot \mathbf{a} = 0$, and its vector component then gives $\mathbf{a} = 0$. Hence $d^2\mathbf{g}/df^2 = 0$, so $\mathbf{g} = c_1 f + c_2$ and the geometric velocity is constant. If $f' = df/dt$ and $\mathbf{g}' = d\mathbf{g}/dt$, then

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}'' f' - \mathbf{g}' f''}{(f')^2} \quad (10.13)$$

$$[U, dU/ds] = 2\gamma^2(\mathbf{v} \times \mathbf{a}) \cdot \mathbf{i}\boldsymbol{\sigma}. \quad (10.14)$$

A Pauli-matrix representation

Throughout this appendix, $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ is a complex paravector and $\mathbf{M} = \text{mat}(Z)$ is its faithful 2×2 matrix image.

A.1 Faithful representation

Every complex element of the displayed algebra basis can be collected as

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (\text{A.1})$$

Here $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Such an element is called a complex paravector. Its faithful 2×2 complex representation begins with

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Thus

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{Bmatrix}. \quad (\text{A.2})$$

It preserves identity, addition, and multiplied order:

$$\text{mat}(1) = \mathbf{I}_2, \quad (\text{A.3})$$

$$\text{mat}(Z'Z) = \text{mat}(Z') \text{mat}(Z). \quad (\text{A.4})$$

A.2 Matrix operations and induced algebra operations

Let $\mathbf{M} = \text{mat}(Z)$ with entries $m_{jk} = a_{jk} + b_{jk}i$, where $a_{jk}, b_{jk} \in \mathbb{R}$. Define transpose by

$$(\mathbf{M}^\text{T})_{jk} := m_{kj}. \quad (\text{A.5})$$

Define entrywise complex conjugation independently by

$$m_{jk}^\text{C} := a_{jk} - b_{jk}i \quad (\text{A.6})$$

$$\mathbf{M}^\text{C} := \begin{Bmatrix} m_{11}^\text{C} & m_{12}^\text{C} \\ m_{21}^\text{C} & m_{22}^\text{C} \end{Bmatrix}. \quad (\text{A.7})$$

The Hermitian transpose is then

$$\mathbf{M}^\text{H} := (\mathbf{M}^\text{C})^\text{T} = (\mathbf{M}^\text{T})^\text{C}. \quad (\text{A.8})$$

Define the algebra operation $^\text{H}$ through the faithful representation by

$$\text{mat}(Z^\text{H}) := \mathbf{M}^\text{H}. \quad (\text{A.9})$$

For $n = 1, 2, 3$, its action on every basis type is

$$\text{mat}(1)^\text{H} = \mathbf{I}_2, \quad (\text{A.10})$$

$$(i\mathbf{I}_2)^\text{H} = -i\mathbf{I}_2, \quad (\text{A.11})$$

$$\text{mat}(\sigma_n)^\text{H} = \text{mat}(\sigma_n), \quad (\text{A.12})$$

$$\text{mat}(i\sigma_n)^\text{H} = -\text{mat}(i\sigma_n). \quad (\text{A.13})$$

Hence $1^\text{H} = 1$, $i^\text{H} = -i$, $\sigma_n^\text{H} = \sigma_n$, and $(i\sigma_n)^\text{H} = -i\sigma_n$. Hermitian conjugation changes matrix-product order. Therefore faithfulness gives

$$\text{mat}((\sigma_n \sigma_m)^\text{H}) = (\text{mat}(\sigma_n) \text{mat}(\sigma_m))^\text{H} \quad (\text{A.14})$$

$$= \text{mat}(\sigma_m)^\text{H} \text{mat}(\sigma_n)^\text{H} = \text{mat}(\sigma_m \sigma_n), \quad (\text{A.15})$$

and hence

$$(\sigma_n \sigma_m)^H = \sigma_m^H \sigma_n^H = \sigma_m \sigma_n. \quad (\text{A.16})$$

Thus H is precisely the Hermitian operator defined in Section 2. To define the distinct sigma conjugation, let $\mathbf{S} := \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}$ and $\mathbf{S}^2 = -\mathbf{I}_2$, so that $\mathbf{S}^{-1} = -\mathbf{S}$, and set

$$\mathbf{M}^* := \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S}, \quad (\text{A.17})$$

$$\text{mat}(\mathbf{Z}^*) := \mathbf{M}^*. \quad (\text{A.18})$$

For $n = 1, 2, 3$, this gives the required action on every basis type:

$$\text{mat}(1)^* = \mathbf{S}^{-1} \mathbf{I}_2^C \mathbf{S} = \mathbf{I}_2, \quad (\text{A.19})$$

$$(\mathbf{i} \mathbf{I}_2)^* = \mathbf{S}^{-1} (\mathbf{i} \mathbf{I}_2)^C \mathbf{S} = -\mathbf{i} \mathbf{I}_2, \quad (\text{A.20})$$

$$\text{mat}(\sigma_n)^* = \mathbf{S}^{-1} \text{mat}(\sigma_n)^C \mathbf{S} = -\text{mat}(\sigma_n), \quad (\text{A.21})$$

$$\text{mat}(\mathbf{i} \sigma_n)^* = \mathbf{S}^{-1} \text{mat}(\mathbf{i} \sigma_n)^C \mathbf{S} = \text{mat}(\mathbf{i} \sigma_n). \quad (\text{A.22})$$

The operation also preserves multiplied order. Since entrywise conjugation preserves order,

$$(\mathbf{M} \mathbf{N})^* = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{N}^C \mathbf{S} \quad (\text{A.23})$$

$$= (\mathbf{S}^{-1} \mathbf{M}^C \mathbf{S}) (\mathbf{S}^{-1} \mathbf{N}^C \mathbf{S}) = \mathbf{M}^* \mathbf{N}^*. \quad (\text{A.24})$$

Therefore

$$(\mathbf{M} \mathbf{N})^* = \mathbf{M}^* \mathbf{N}^*. \quad (\text{A.25})$$

The four operations must remain distinct:

symbol	matrix operation	relation to the algebra
$\mathbf{M}^T$	transpose	changes matrix-product order
$\mathbf{M}^C$	entrywise complex conjugation	sends $\mathbf{i}$ to $-\mathbf{i}$
$\mathbf{M}^H$	Hermitian transpose	$\text{mat}(\mathbf{Z}^H)$
$\mathbf{M}^*$	sigma conjugation	$\text{mat}(\mathbf{Z}^*)$

Their product-order rules are

$$(\mathbf{M} \mathbf{N})^T = \mathbf{N}^T \mathbf{M}^T, \quad (\text{A.26})$$

$$(\mathbf{M} \mathbf{N})^C = \mathbf{M}^C \mathbf{N}^C, \quad (\text{A.27})$$

$$(\mathbf{M} \mathbf{N})^H = \mathbf{N}^H \mathbf{M}^H. \quad (\text{A.28})$$

Sigma conjugation preserves order by equation (A.25). Thus the induced algebra operations satisfy

$$(\mathbf{Z}^H)^H = \mathbf{Z}, \quad (\text{A.29})$$

$$(\mathbf{Z} \mathbf{Z}')^H = \mathbf{Z}'^H \mathbf{Z}^H, \quad (\text{A.30})$$

$$(\mathbf{Z}^*)^* = \mathbf{Z}, \quad (\text{A.31})$$

$$(\mathbf{Z} \mathbf{Z}')^* = \mathbf{Z}^* \mathbf{Z}'^*, \quad (\text{A.32})$$

$$\mathbf{Z}^{*H} = \mathbf{Z}^{H*}. \quad (\text{A.33})$$

A.2.1 Trace, determinant, and adjugate

Recall $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $\mathbf{M} = \text{mat}(Z)$. Define the matrix trace and determinant by

$$\text{tr}(\mathbf{M}) := m_{11} + m_{22} \quad (\text{A.34})$$

$$\det(\mathbf{M}) := m_{11}m_{22} - m_{12}m_{21}. \quad (\text{A.35})$$

The corresponding paravector operations are

$$\text{tr}(Z) := \text{tr}(\mathbf{M}) = 2S, \quad (\text{A.36})$$

$$\det(Z) := \det(\mathbf{M}). \quad (\text{A.37})$$

Faithfulness transfers the matrix identities to the algebra:

$$\text{tr}(Z + Z') = \text{tr}(Z) + \text{tr}(Z'), \quad (\text{A.38})$$

$$\text{tr}(cZ) = c \text{tr}(Z), \quad (\text{A.39})$$

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (\text{A.40})$$

$$\det(ZZ') = \det(Z) \det(Z'), \quad (\text{A.41})$$

$$\det(cZ) = c^2 \det(Z). \quad (\text{A.42})$$

The last identity holds for $c \in \mathbb{C}$. For a 2×2 matrix and its paravector preimage, define

$$\text{adj}(\mathbf{M}) := \text{tr}(\mathbf{M})\mathbf{I}_2 - \mathbf{M}, \quad (\text{A.43})$$

$$\text{mat}(\text{adj}(Z)) := \text{adj}(\mathbf{M}). \quad (\text{A.44})$$

Direct comparison with the induced algebra operators gives

$$\text{adj}(Z) = Z^{*\text{H}} \quad (\text{A.45})$$

$$\text{mat}(\text{adj}(Z)) = \text{adj}(\mathbf{M}) = (\mathbf{M}^*)^{\text{H}}. \quad (\text{A.46})$$

Equivalently,

$$\text{adj}(Z) = \text{tr}(Z) - Z \quad (\text{A.47})$$

$$\text{tr}(Z) = Z + \text{adj}(Z) = 2S. \quad (\text{A.48})$$

The adjugate changes product order and obeys

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z), \quad (\text{A.49})$$

$$\text{adj}(\text{adj}(Z)) = Z, \quad (\text{A.50})$$

$$\text{adj}(Z) Z = Z \text{adj}(Z) = \det(Z), \quad (\text{A.51})$$

$$\text{mat}(\text{adj}(Z) Z) = \det(\mathbf{M})\mathbf{I}_2. \quad (\text{A.52})$$

Consequently, when $\det(Z) \neq 0$,

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)}. \quad (\text{A.53})$$

Finally, Cayley–Hamilton gives

$$\mathbf{M}^2 - \text{tr}(\mathbf{M})\mathbf{M} + \det(\mathbf{M})\mathbf{I}_2 = 0, \quad (\text{A.54})$$

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (\text{A.55})$$

A.3 Algebra–matrix crosswalk

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$\text{sc}(Z) := S \quad (\text{A.56})$$

$$\text{vec}(Z) := \mathbf{V}. \quad (\text{A.57})$$

Thus $\text{vec}(Z)$ is the coefficient vector, while its algebra element is $\mathbf{V} \cdot \boldsymbol{\sigma}$.

For $\mathbf{M} = \text{mat}(Z)$, the extraction formulas are

$$\text{sc}(Z) = S = \frac{1}{2} \text{tr}(\mathbf{M}), \quad (\text{A.58})$$

$$\text{mat}(\text{sc}(Z)) = S \mathbf{I}_2, \quad (\text{A.59})$$

$$(\text{vec}(Z))_n = V_n = \frac{1}{2} \text{tr}(\boldsymbol{\sigma}_n \mathbf{M}), \quad (\text{A.60})$$

$$\text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) = \mathbf{M} - \frac{1}{2} \text{tr}(\mathbf{M}) \mathbf{I}_2. \quad (\text{A.61})$$

Define real and imaginary paravector extraction through Hermitian splitting:

$$\text{mat}(\text{re}(Z)) := \frac{1}{2}(\mathbf{M} + \mathbf{M}^{\text{H}}), \quad (\text{A.62})$$

$$\text{mat}(\text{im}(Z)) := \frac{\mathbf{M} - \mathbf{M}^{\text{H}}}{2i}. \quad (\text{A.63})$$

By construction, the algebra and matrix operators agree through

$$\text{mat}(Z^{\text{H}}) = \mathbf{M}^{\text{H}} \quad (\text{A.64})$$

$$\text{mat}(Z^*) = \mathbf{M}^* \quad (\text{A.65})$$

$$\text{mat}(Z^{*\text{H}}) = (\mathbf{M}^*)^{\text{H}}. \quad (\text{A.66})$$

The matrix-only operations $^{\text{C}}$ and $^{\text{T}}$ are related to the algebra operations by

$$\mathbf{M}^{\text{C}} = \mathbf{S} \mathbf{M}^* \mathbf{S}^{-1} \quad (\text{A.67})$$

$$\mathbf{M}^{\text{T}} = \mathbf{S} \text{mat}(Z^{*\text{H}}) \mathbf{S}^{-1}. \quad (\text{A.68})$$

Finally, scalar extraction of a product and the paravector norm are

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}_{Z'} \mathbf{M}_Z), \quad (\text{A.69})$$

$$\|Z\|^2 := \frac{1}{2} \text{tr}(\mathbf{M}^{\text{H}} \mathbf{M}) = \frac{1}{2} \text{tr}(\mathbf{M} \mathbf{M}^{\text{H}}). \quad (\text{A.70})$$

A.4 Matrix amplitudes and column spinors

Let ψ and ψ' be paravector amplitudes and Z a paravector observable, and define

$$\mathbf{M}_{\psi} := \text{mat}(\psi) \quad (\text{A.71})$$

$$\mathbf{M}_{\psi'} := \text{mat}(\psi'). \quad (\text{A.72})$$

This subsection translates the intrinsic paravector spinor development of Section 8.2 into the familiar matrix and column languages. Faithfulness and Hermitian compatibility give

$$\text{mat}(\psi'^{\text{H}} Z \psi) = \mathbf{M}_{\psi'}^{\text{H}} \text{mat}(Z) \mathbf{M}_{\psi}. \quad (\text{A.73})$$

Scalar extraction is half the matrix trace, so the intrinsic definitions (8.1) and (8.2) become

$$\langle \psi' | \psi \rangle = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_\psi), \quad (\text{A.74})$$

$$\langle \psi' | Z | \psi \rangle = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_\psi). \quad (\text{A.75})$$

Thus $\text{mat}(\psi^H \psi) = \mathbf{M}_\psi^H \mathbf{M}_\psi$ is generally Hermitian rather than scalar; the scalar inner product is obtained by taking half its trace.

An unrestricted matrix $\mathbf{M}_\psi$ contains two independent columns. Under left multiplication those columns transform as two equivalent two-component carriers, so the full matrix is not yet one conventional Pauli column. The fixed right projection below selects one carrier.

Recall the fixed idempotent from Section 8.2:

$$\Pi_3 = \frac{1}{2}(1 + \sigma_3). \quad (\text{A.76})$$

Here Π_3 is the fixed column selector and is distinct from the matrix superscript C used for entrywise complex conjugation. With

$$\mathbf{v} := \{1, 0\}^T \quad (\text{A.77})$$

$$\mathbf{v}^H \mathbf{v} = 1, \quad (\text{A.78})$$

its matrix image is

$$\text{mat}(\Pi_3) = \mathbf{v} \mathbf{v}^H = \begin{Bmatrix} 1 & 0 \\ 0 & 0 \end{Bmatrix}. \quad (\text{A.79})$$

Any other fixed primitive Hermitian idempotent gives an equivalent column basis, but Π_3 is kept fixed throughout the following formulas.

For unrestricted complex paravectors Z and Z' , let $\psi = Z\Pi_3$ and $\psi' = Z'\Pi_3$ be two ideal amplitudes. Define their conventional columns by

$$\psi := \frac{1}{\sqrt{2}} \mathbf{M}_\psi \mathbf{v} \quad (\text{A.80})$$

$$\psi' := \frac{1}{\sqrt{2}} \mathbf{M}_{\psi'} \mathbf{v}. \quad (\text{A.81})$$

The inverse relations are

$$\mathbf{M}_\psi = \sqrt{2} \psi \mathbf{v}^H \quad (\text{A.82})$$

$$\mathbf{M}_{\psi'} = \sqrt{2} \psi' \mathbf{v}^H. \quad (\text{A.83})$$

For the normalized ideal basis in equation (8.13), this map gives explicitly

$$\frac{1}{\sqrt{2}} \text{mat}(e_+) \mathbf{v} = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}, \quad \frac{1}{\sqrt{2}} \text{mat}(e_-) \mathbf{v} = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}. \quad (\text{A.84})$$

Thus each ideal amplitude and its column contain the same two complex components. Its matrix has rank one for a nonzero state and rank zero for the zero state. The fixed right factor records the chosen ideal, while all state information is contained in the column.

Substitution into the trace pairing gives exactly the conventional formulas:

$$\langle \psi' | \psi \rangle = \psi'^H \psi \quad (\text{A.85})$$

$$\langle \psi' | Z | \psi \rangle = \psi'^H \text{mat}(Z) \psi. \quad (\text{A.86})$$

In particular,

$$\text{sc}(\psi^H \psi) = \psi^H \psi \quad (\text{A.87})$$

$$\langle Z \rangle_\psi = \frac{\psi^H \text{mat}(Z) \psi}{\psi^H \psi}. \quad (\text{A.88})$$

The factor $1/\sqrt{2}$ in the column map compensates for scalar extraction being one half of the matrix trace. Left multiplication needs no further adjustment: $T\psi$ maps to $\text{mat}(T)\psi$. In particular, the rotor law $\psi' = R^H \psi$ becomes

$$\psi' = \text{mat}(R)^H \psi, \quad (\text{A.89})$$

which rotates the column and its Bloch vector in the same direction as the Rodrigues formula in Section 9.

For a measurement along a unit direction $\mathbf{u}$, let

$$\Pi_\pm = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \psi_\pm = \Pi_\pm \psi. \quad (\text{A.90})$$

The corresponding matrices and columns are

$$\mathbf{M}_{\psi_\pm} = \text{mat}(\Pi_\pm) \mathbf{M}_\psi, \quad (\text{A.91})$$

$$\psi_\pm = \text{mat}(\Pi_\pm) \psi. \quad (\text{A.92})$$

Consequently, the probability in equation (8.27) is

$$p_\pm(\mathbf{u}) = \frac{\psi_\pm^H \psi_\pm}{\psi^H \psi} \quad (\text{A.93})$$

$$= \frac{\psi^H \text{mat}(\Pi_\pm) \psi}{\psi^H \psi} \quad (\text{A.94})$$

$$= \frac{\text{tr}(\mathbf{M}_\psi^H \text{mat}(\Pi_\pm) \mathbf{M}_\psi)}{\text{tr}(\mathbf{M}_\psi^H \mathbf{M}_\psi)}. \quad (\text{A.95})$$

For an ideal amplitude, its normalized density matrix may therefore be written in any of the equivalent forms

$$\frac{\psi \psi^H}{\psi^H \psi} = \frac{\mathbf{M}_\psi \mathbf{M}_\psi^H}{\text{tr}(\mathbf{M}_\psi \mathbf{M}_\psi^H)} = \frac{1}{2} \text{mat}(B). \quad (\text{A.96})$$

For an unrestricted matrix amplitude, the matrix-amplitude expression remains a valid normalized positive density matrix and may describe a mixed state. It comes from a single Pauli column precisely when the matrix amplitude has rank one.

For the Dirac block, apply the same column map to each ideal entry and write ψ_1, ψ_2 for the resulting columns. The faithful block action is

$$\mathbf{W}(Z) \Psi \longleftrightarrow \left\{ \begin{array}{cc} 0 & \text{mat}(Z) \\ \text{mat}(\text{adj}(Z)) & 0 \end{array} \right\} \left\{ \begin{array}{c} \psi_1 \\ \psi_2 \end{array} \right\}. \quad (\text{A.97})$$

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, this block matrix is

$$\left\{ \begin{array}{cc} 0 & S\mathbf{I}_2 + \mathbf{V} \cdot \boldsymbol{\sigma} \\ S\mathbf{I}_2 - \mathbf{V} \cdot \boldsymbol{\sigma} & 0 \end{array} \right\}, \quad (\text{A.98})$$

which is the standard chiral form used by the Dirac equation. In this ordering the conventional Hermitian row entering the Dirac current is $\{\psi_2^H, \psi_1^H\}$. The normalized spin state in equation (8.114) becomes

$$\frac{1}{2} \text{mat}(B) = \frac{\psi_1 \psi_1^H + \psi_2 \psi_2^H}{\psi_1^H \psi_1 + \psi_2^H \psi_2}. \quad (\text{A.99})$$

The charge-reversal rule also has a direct column form. Entrywise conjugation of a column is denoted by the already defined matrix superscript C . The ideal map gives

$$\frac{1}{\sqrt{2}} \text{mat}(\psi^* \sigma_1) \mathbf{v} = \mathbf{S}^{-1} \psi^C, \quad (\text{A.100})$$

so equation (8.129) becomes

$$\Psi_c \longleftrightarrow \begin{Bmatrix} \mathbf{S}^{-1} \psi_2^C \\ -\mathbf{S}^{-1} \psi_1^C \end{Bmatrix}. \quad (\text{A.101})$$

This is the familiar antisymmetric two-component charge-reversal map in the chiral ordering used here. By contrast, the matrix image of equation (8.75) is

$$\text{mat}(J) = (\psi_1^H \psi_1 + \psi_2^H \psi_2) \mathbf{I}_2 \quad (\text{A.102})$$

$$+ \sum_{n=1}^3 \left(-\psi_1^H \sigma_n \psi_1 + \psi_2^H \sigma_n \psi_2 \right) \sigma_n. \quad (\text{A.103})$$

The rotation and boost laws of the two columns are correspondingly

$$\begin{Bmatrix} \psi'_1 \\ \psi'_2 \end{Bmatrix} = \begin{Bmatrix} \text{mat}(R)^H & 0 \\ 0 & \text{mat}(R)^H \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \end{Bmatrix}, \quad (\text{A.104})$$

$$\begin{Bmatrix} \psi'_1 \\ \psi'_2 \end{Bmatrix} = \begin{Bmatrix} \text{mat}(L) & 0 \\ 0 & \text{mat}(L)^{-1} \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \end{Bmatrix}. \quad (\text{A.105})$$

Thus the two columns have the same rotation generator and opposite boost generators, which is the standard chiral Dirac transformation.

A.5 Quaternion subalgebra

The real span of the scalar and bivector basis elements is a quaternion subalgebra of the complex paravectors. Define its three imaginary units by

$$q_1 = -\sigma_2 \sigma_3 = -i\sigma_1 \quad (\text{A.106})$$

$$q_2 = -\sigma_3 \sigma_1 = -i\sigma_2 \quad (\text{A.107})$$

$$q_3 = -\sigma_1 \sigma_2 = -i\sigma_3, \quad (\text{A.108})$$

or, collectively, $\mathbf{q} = -i\boldsymbol{\sigma}$. For $n, m \in \{1, 2, 3\}$, the sigma multiplication rule gives

$$q_n q_m = -\delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} q_k, \quad (\text{A.109})$$

Consequently,

$$q_1^2 = q_2^2 = q_3^2 = -1 \quad (\text{A.110})$$

$$q_1 q_2 = q_3 \quad (\text{A.111})$$

$$q_1 q_2 q_3 = -1, \quad (\text{A.112})$$

with the remaining products fixed by anticommutation. For $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$, define the real paravector

$$Q = a + \mathbf{A} \cdot \mathbf{q} = a - \mathbf{iA} \cdot \boldsymbol{\sigma}. \quad (\text{A.113})$$

These are precisely the quaternions embedded in the paravector algebra. For a second quaternion $Q' = a' + \mathbf{A}' \cdot \mathbf{q}$, equation (A.109) gives the familiar product

$$QQ' = (aa' - \mathbf{A} \cdot \mathbf{A}') + (a\mathbf{A}' + a'\mathbf{A} + \mathbf{A} \times \mathbf{A}') \cdot \mathbf{q}. \quad (\text{A.114})$$

Hermitian conjugation reverses every bivector, while sigma conjugation fixes it:

$$q_n^{\text{H}} = -q_n \quad (\text{A.115})$$

$$q_n^* = q_n. \quad (\text{A.116})$$

Quaternion conjugation is therefore already the Hermitian operation,

$$Q^{\text{H}} = a - \mathbf{A} \cdot \mathbf{q} \quad (\text{A.117})$$

$$Q^* = Q \quad (\text{A.118})$$

$$\text{adj}(Q) = Q^{\text{H}}. \quad (\text{A.119})$$

It follows that

$$QQ^{\text{H}} = Q^{\text{H}}Q = a^2 + \mathbf{A}^2 = \det(Q) \quad (\text{A.120})$$

$$Q^{-1} = \frac{Q^{\text{H}}}{a^2 + \mathbf{A}^2} \quad (\text{A.121})$$

for nonzero Q .

The faithful matrix representation sends the quaternion units to

$$\text{mat}(q_1) = \begin{Bmatrix} 0 & -\mathbf{i} \\ -\mathbf{i} & 0 \end{Bmatrix} \quad (\text{A.122})$$

$$\text{mat}(q_2) = \begin{Bmatrix} 0 & -1 \\ 1 & 0 \end{Bmatrix} \quad (\text{A.123})$$

$$\text{mat}(q_3) = \begin{Bmatrix} -\mathbf{i} & 0 \\ 0 & \mathbf{i} \end{Bmatrix}. \quad (\text{A.124})$$

Hence equation (A.113) becomes

$$\text{mat}(Q) = a\mathbf{I}_2 - \mathbf{iA} \cdot \boldsymbol{\sigma} = \begin{Bmatrix} a - \mathbf{iA}_3 & -A_2 - \mathbf{iA}_1 \\ A_2 - \mathbf{iA}_1 & a + \mathbf{iA}_3 \end{Bmatrix}. \quad (\text{A.125})$$

The algebraic conjugation and norm become the usual matrix identities

$$\text{mat}(Q^{\text{H}}) = \text{mat}(Q)^{\text{H}}, \quad (\text{A.126})$$

$$\text{mat}(Q)^{\text{H}} \text{mat}(Q) = \text{mat}(Q) \text{mat}(Q)^{\text{H}} = (a^2 + \mathbf{A}^2)\mathbf{I}_2, \quad (\text{A.127})$$

$$\det(\text{mat}(Q)) = a^2 + \mathbf{A}^2. \quad (\text{A.128})$$

Thus a unit quaternion is represented by a unitary matrix of determinant one.

Finally, for a unit axis $\mathbf{u}$, let

$$Q = \cos \frac{\theta}{2} + \mathbf{u} \cdot \mathbf{q} \sin \frac{\theta}{2}. \quad (\text{A.129})$$

Since $\mathbf{q} = -\mathbf{i}\boldsymbol{\sigma}$, this quaternion is R^{H} for the rotor R defined in Section 9. The quaternion sandwich

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^{\text{H}} \quad (\text{A.130})$$

therefore reproduces the same Rodrigues rotation.

B Reference summary

This index collects the principal definitions and results in natural units, with signature $(+, -, -, -)$. Topic titles and equation numbers link to their full statements and derivations.

Algebra and vector operations.

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k \quad (1.9),$$

$$[\sigma_n, \sigma_m] = 2 \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k \quad (1.9),$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma},$$

$$\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A},$$

$$\|\mathbf{Z}\|^2 = \|\mathbf{S}\|^2 + \|\mathbf{V}\|^2,$$

$$\|\mathbf{V}\|^2 = \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} \quad (3.87).$$

Hermitian and sigma-conjugate operators.

$$(ZZ')^* = Z^* Z'^*,$$

$$(ZZ')^H = Z'^H Z^H,$$

$$\text{adj}(Z) = Z^{*H},$$

$$\text{adj}(ZZ') = Z'^{*H} Z^{*H} \quad (3.13).$$

Paravectors, extraction, and products. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = (a + bi) + (\mathbf{A} + B\mathbf{i}) \cdot \boldsymbol{\sigma}$,

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*H}),$$

$$\mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*H}),$$

$$a = \text{sc}(\text{re}(Z)),$$

$$b = \text{sc}(\text{im}(Z)),$$

$$\mathbf{A} = \text{vec}(\text{re}(Z)),$$

$$\mathbf{B} = \text{vec}(\text{im}(Z)),$$

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma},$$

$$\det(Z) = \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2,$$

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)},$$

$$UU^{-1} = U^{-1}U = 1 \quad (3.24),$$

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0,$$

$$\lambda_{\pm} = S \pm \sqrt{\mathbf{V}^2}.$$

See the product (3.70), determinant (3.15), and quadratic identity (3.57).

Projection operators and functions.

$$\begin{aligned}
\Pi_{\pm} &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
\Pi_{\pm}^2 &= \Pi_{\pm}, \\
\Pi_+ \Pi_- &= 0, \\
f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_-, \\
X &= \lambda_+ \Pi_+ + \lambda_- \Pi_-, \\
f(X) &= f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_-.
\end{aligned}$$

See the general complex-paravector rule (3.166) and ladder operators in Section 3.6.1.

Paravector differential operators.

$$\begin{aligned}
\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\
\partial^* &= \text{adj}(\partial) = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\
\Box &= \partial\partial^* = \partial^*\partial = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}
\end{aligned} \tag{5.7}$$

Space–time kinematics and action.

$$\begin{aligned}
dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\
ds^2 &= dX dX^* = dt^2 - d\mathbf{r}^2 \\
\gamma &= (1 - \mathbf{v}^2)^{-1/2}, \\
ds &= dt/\gamma, \\
\frac{d}{ds} &= \gamma \frac{d}{dt}, \\
U &= dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) \\
P &= mU = E + \mathbf{p} \cdot \boldsymbol{\sigma} \\
E &= \text{sc}(P), \\
E^2 - \mathbf{p}^2 &= m^2, \\
\int L dt &= -m \int ds, \\
L &= -m/\gamma, \\
H &= m\gamma, \\
K &= H - m = m(\gamma - 1)
\end{aligned} \tag{6.3}, \tag{6.22}, \tag{6.26}, \tag{6.70}.$$

Electromagnetism.

$$\begin{aligned}
F &= (\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, \\
D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma}, \\
\partial F &= D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma} \\
\partial \Phi &= S + F^*, \\
\mathbf{E} &= -\partial_t \mathbf{A} - \partial_r V, \\
\mathbf{B} &= \partial_r \times \mathbf{A}, \\
\rho &= \square V - \partial_t S \\
\mathbf{J} &= \square \mathbf{A} + \partial_r S, \\
\frac{1}{2} F^2 &= \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) + \mathbf{i} \mathbf{E} \cdot \mathbf{B}, \\
\frac{1}{2} F F^H &= \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
T(N) &= \frac{1}{2} F N F^H \\
\frac{dP}{ds} &= m \frac{dU}{ds} = q \operatorname{re}(FU), \\
\frac{d\mathbf{p}}{dt} &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \\
\mathbf{a} &= \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v})
\end{aligned}
\tag{7.4}, \tag{7.28}, \tag{7.91}, \tag{7.48}, \tag{7.82}.$$

See the field invariants (7.44) and Poynting balance (7.101). The connection to invariant proper acceleration is developed in Section 7.4.

Wave equations, spin, and energy sectors. Here Π_3 is the fixed ideal selector, while $\Pi_{\pm}$ are measurement projectors. In the energy formulas, H denotes the free Dirac Hamiltonian. The unintegrated ratios below apply to finite amplitudes and pointwise local states; for spatial

wavefunctions, integrate the scalar numerator and denominator before taking their ratio.

$$\begin{aligned}
\langle \psi' | \psi \rangle &= \text{sc}(\psi'^H \psi), \\
\langle \psi' | Z | \psi \rangle &= \text{sc}(\psi'^H Z \psi), \\
\langle Z \rangle_\psi &= \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^H Z \psi)}{\text{sc}(\psi^H \psi)}, \\
\Pi_3 &= \frac{1}{2}(1 + \sigma_3), & \psi \Pi_3 &= \psi, \\
B &= \frac{\psi \psi^H}{\text{sc}(\psi^H \psi)} = 1 + \mathbf{s} \cdot \boldsymbol{\sigma}, \\
p_\pm(\mathbf{u}) &= \text{sc}(\Pi_\pm B) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}), \\
P &= E + \mathbf{p} \cdot \boldsymbol{\sigma}, & w &= \exp(-i \text{sc}(\text{adj}(P) X)), \\
i\partial^* w &= Pw, & i\partial w &= \text{adj}(P) w, \\
(\square + m^2)\psi &= 0, \\
\mathbf{W}(Z) &= \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix}, \\
(\mathbf{W}(i\partial) - m\mathbf{I})\Psi &= 0, \\
(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi &= 0, \\
i\partial_t \psi &= \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \psi, \\
J &= \text{adj}(\Psi_1 \Psi_1^H) + \Psi_2 \Psi_2^H & (8.75), \\
\Pi_\pm &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
U &= P/m = (E + \mathbf{p} \cdot \boldsymbol{\sigma})/m, & \det(U) &= 1 \quad (m > 0, E > 0), \\
H &= \begin{Bmatrix} -\mathbf{p} \cdot \boldsymbol{\sigma} & m \\ m & \mathbf{p} \cdot \boldsymbol{\sigma} \end{Bmatrix}, & E &= \sqrt{\mathbf{p}^2 + m^2}, \\
Q_\pm &= \frac{1}{2}(\mathbf{I} \pm H/E) & (8.120), \\
\text{spin-up projection:} & \Pi_+ \Psi, \\
\text{spin-down projection:} & \Pi_- \Psi, \\
\Psi_c &= \{\Psi_2^* \sigma_1, -\Psi_1^* \sigma_1\}^\top & (8.129).
\end{aligned}$$

See the paravector inner product (8.1), minimal-ideal development in Section 8.2, spin-carrier comparison in Section 8.3, Klein–Gordon equation (8.49), Dirac equations (8.67) and (8.73), general Klein–Gordon lift (8.72), Pauli equation (8.89), spin probability (8.27), energy probabilities (8.121), and joint rule (8.131). The covariant mass-shell idempotents $\frac{1}{2}(\mathbf{I} \pm \mathbf{W}(\text{adj}(U)))$ are distinct from the Hermitian energy projectors summarized here.

Rotations and boosts.

$$\begin{aligned}
R &= \exp(\mathrm{i}\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \\
R^{\mathrm{H}} &= R^{-1}, \\
Z' &= R^{\mathrm{H}} Z R, \\
\psi' &= R^{\mathrm{H}} \psi, \\
L &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \\
L^{\mathrm{H}} &= L, \\
X' &= L^{-1} X L^{-1}, \\
\Psi'_1 &= L \Psi_1, & \Psi'_2 &= L^{-1} \Psi_2, \\
\partial' &= L \partial L, & D' &= L^{-1} D L^{-1}, \\
F' &= L^{-1} F L, & J' &= L^{-1} J L^{-1}.
\end{aligned}$$

See the bilinear rule (8.8), Dirac boost rule (9.13), and varying-frame derivatives in Section 6.2.

Matrix representation in Appendix A.

$$\mathrm{mat}(Z) = \begin{Bmatrix} S + V_3 & V_1 - \mathrm{i}V_2 \\ V_1 + \mathrm{i}V_2 & S - V_3 \end{Bmatrix} \quad (\text{A.2}),$$

$$\mathrm{mat}(ZZ') = \mathrm{mat}(Z) \mathrm{mat}(Z') \quad (\text{A.4}),$$

$$\mathrm{mat}(Z^{\mathrm{H}}) = \mathrm{mat}(Z)^{\mathrm{H}},$$

$$\mathrm{mat}(Z^*) = \mathbf{S}^{-1} \mathrm{mat}(Z)^{\mathrm{C}} \mathbf{S} \quad (\text{A.18}),$$

$$\mathrm{mat}(\mathrm{adj}(Z)) = \mathrm{adj}(\mathrm{mat}(Z)),$$

$$\det(Z) = \det(\mathrm{mat}(Z)) \quad (\text{A.52}).$$

The minimal-ideal column map is equation (A.80).

Quaternion subalgebra in Appendix A.

$$\begin{aligned}
\mathbf{q} &= -\mathrm{i}\boldsymbol{\sigma}, \\
Q^{\mathrm{H}} &= a - \mathbf{A} \cdot \mathbf{q}, \\
QQ^{\mathrm{H}} &= a^2 + \mathbf{A}^2.
\end{aligned}$$

See the quaternion product (A.109), its paravector embedding (A.113), and its faithful matrix form (A.125).

Flat spacetime in generalized coordinates.

$$\begin{aligned}
X &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, \\
\mathrm{d}s^2 &= \mathrm{d}f^2 - \mathrm{d}\mathbf{g}^2, \\
\mathbf{v} &= \mathrm{d}\mathbf{g}/\mathrm{d}f, \\
U &= \mathrm{d}X/\mathrm{d}s = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).
\end{aligned}$$

C Conventions and cautions

- The same symbol $i = \sigma_1\sigma_2\sigma_3$ is both the algebra's pseudoscalar and, in the Pauli representation, the ordinary complex unit.
- Sigma conjugation $(\cdot)^*$, the Hermitian operator $(\cdot)^H$, and their composition are distinct. In the Pauli representation, the Hermitian operator corresponds to Hermitian adjunction.
- $\det(Z)$ is an indefinite quadratic form and may be complex for complex Z .
- Differential operators act on everything to their right unless parentheses state otherwise. This matters in products involving fields.
- Vectors and matrices are bold, and only vectors and matrices are bold. In particular, the electromagnetic fields formerly written as calligraphic or italic E and B are the upright bold vectors $\mathbf{E}$ and $\mathbf{B}$, while scalar energy remains the unbold symbol E . Blackboard bold is retained only for the number sets $\mathbb{R}$ and $\mathbb{C}$.
- A unit paravector U is defined by the existence of a two-sided inverse, $UU^{-1} = U^{-1}U = 1$. In particle applications the local recall $U = dX/ds$ specializes this general notation to proper velocity; that real unit additionally satisfies $\det(U) = 1$ and $U^{-1} = U^*$.
- The action is written as $\int L dt$, avoiding conflict with the scalar part S . The Lagrangian L and Hamiltonian H use the regular math font. The Dirac block function is the upright bold $\mathbf{W}(\cdot)$, and a vector square is explicitly defined by $\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}$.
- Curly braces delimit explicit vector-entry lists and matrices, while commutators alone use square brackets. Parentheses delimit functions, algebraic grouping, and order-sensitive operator products. Redundant grouping parentheses were removed.
- The extraction functions $\text{sc}(\cdot)$, $\text{vec}(\cdot)$, $\text{re}(\cdot)$, and $\text{im}(\cdot)$ replace `scalar`, `vector`, and their capitalized variants. They are defined before use. The unused expectation abbreviation `Exp` was removed.
- The original metric notation $\langle\langle Z \rangle\rangle^2$ and determinant bars were replaced by $\det(Z)$. Double bars are reserved for squared norms, with

$$\|Z\|^2 = \|S\|^2 + \|\mathbf{V}\|^2.$$

Absolute-value bars are not used for paravector norms. Entrywise complex-conjugation notation is confined to the matrix appendix.

- Sigma conjugation retains the splat $*$. The operation originally written R , and elsewhere as a dagger, is written H because it reverses multiplied sigma order and becomes Hermitian adjunction in the matrix representation. The abbreviation $\text{adj}(Z) := Z^{*H}$ is defined early and used where it shortens formulas.
- The matrix appendix uses matching upright sans-serif capitals T , C , and H for transpose, entrywise complex conjugation, and Hermitian transpose. It distinguishes these operations from sigma conjugation and states which operations reverse product order.
- The fixed idempotent $\Pi_3 = \frac{1}{2}(1 + \sigma_3)$ is multiplied on the right to select one minimal left ideal and one Pauli column. The variable projectors $\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$ act from the left and represent physical spin measurements. When $\mathbf{u}$ is the third-axis direction, Π_3 and Π_+ are the same algebra element, but right ideal selection and left measurement remain different actions.
- The adjugate–determinant identity applies directly to algebraic paravectors and to free differential paravectors whose coefficient operators commute. Gauge-covariant energy and momentum components do not commute; their products remain ordered and their vector remainders are the electromagnetic field couplings.
- On Dirac blocks, H transposes the block and applies the same algebraic H to every entry, while

- * acts entrywise without transposing the block. For spatial differential operators, Hermiticity additionally includes the operator domain, boundary conditions, and integration by parts.
- Plus/minus labels are subscripts ($\pm$, $+$, $-$), not superscripts or parenthesized labels. Parallel and perpendicular components use $\parallel$ and $\perp$ rather than **par** and **prp**; their projection formulas are restated where needed.
 - Spatial derivatives use ∂_r rather than ∇ , and no overbars are used. Literal or malformed source commands such as **quad** and **frac** were converted to formatted L^AT_EX.
 - Long product expansions were consolidated while retaining their distinct components and results. Expressions otherwise preserve the compact ordering of the source, for example $\text{sc}(\Phi^*U)$ with $U = \text{d}X/\text{d}s$, without an extra pair of grouping parentheses.